Alternative Shared Kitchen

Temitope Akintade

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Examiner: Professor Yvonne Eriksson
Supervisor: Dr. Senem Turhan
The School of Innovation, Design and Engineering
Mälardalen University
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Mälardalens universitet

Abstract

This master's thesis aims to contribute to the design and innovation management literature by exploring to understand an alternative shared social kitchen that is people-driven around shared interests, promoting social well-being. This is in order to explore the potential of systemic, regenerative, speculative sustainable design in addressing challenges faced by shared kitchens, while understanding diverse user needs within enabling contexts. Through a design process of ethnography research, frameworks, imperatives and solution finding, the kitchen was explored from different participants' perspectives and context. The resulting scenario from the ethnography research highlighted technology, food waste, circularity and food system scenarios. The study contributes to future dialogue and understanding of the different processes and perspectives in imaginative shared kitchen using an explorative approach. It is vital to understand the holistic picture of the shared kitchen and not only focus on the functional, environmental and economic aspect, but also on the social development for a more inclusive design.

Keywords: Shared kitchen, scenarios, speculative, regenerative, innovation, systemic, artifact, neighbourhood

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1.1 **Introduction**

An increasing number of people are gathering in cities around the globe. People's quality of life and health are significantly impacted by the way major urban centres are designed, particularly as population density increases. This brings both possibilities and challenges: although many urban environments encourage inactivity and loneliness, there is also potential for health-promoting policies and programmes to have a significant influence in densely populated regions. Juul (2023) emphasised that we should view well-being holistically as including the needs of our physical, social, and mental selves if we are to account for the complexity of the variables that influence it. In the city, we must make room for everyone, including those who are different from us. Communities cannot be created by one person acting alone; organisations must assure community hosting through common co-creation interests (Juul, 2023). Public goods, including ecological and social concerns, biodiversity, and ecosystem services, among other sustainability issues, are increasingly being governed by organisations (D'Amato et al., 2022). Both governmental bodies (such as the United Nations Rio+20 and Agenda 2030 processes) and academics have argued for the need of private sector engagement in the co-governance of socio-ecological concerns (Addison et al., 2019; Barbier et al., 2018; Whiteman et al., 2013). Increased firm engagement in socially responsible practices has resulted from a growing level of activism and commitment to environmental, social, and governance practices. This has also led to greater firm disclosure about these practices to increase stakeholder awareness (Pulino et al., 2022). Urban sustainability is used to include many aspects of city planning, as pointed out by Terama et al. (2019). In response to both national and international sustainable development goal (SDG) objectives, sustainability has been used liberally to promote goals that are both ubiquitous and hard to quantify. The processes of controlled and ad hoc urbanisation vary, and there are local differences in how sustainability is defined. Although green city planning principles are leading the way in sustainability and improving sustainable performance, important sustainability challenges, particularly in the social domain remain unaddressed (Terama et al., 2019). The effectiveness of the practical measures depends on one another due to the interconnectivity of social, environmental, economic, and additional sustainability factors. Therefore, it is crucial to implement transformative urban sustainability in accordance with the SDG framework.

New buildings must reduce their carbon content by 94% by 2050 to satisfy Paris Climate Agreement commitments (Kertsmik et al., 2023). According to the New Bauhaus, when faced with material constraints, we must find new immaterial abundances. A limited material future will force us to accelerate civic economy development that encourages radical sharing and creates new abundant paths. We will need to develop new regenerative interactions with nature, soil, water, space, and time in this future. To survive natural disasters like climate change and the COVID-19 pandemic, food supply systems must be resilient to socioeconomic and geopolitical disruptions. The dependence of urban populations on external food sources renders them susceptible to systemic failures (Jin & Zhu, 2023). Hybrid models, zero waste, circularity, sharing scarce key resources, and regenerative, bio-based, and non-invasive materials could form part of a new economy. Such an economy could be centred

on care, stewardship, radical creativity, and innovation rather than ownership and control. The proliferation of technologies has also created a cultural and behavioural disconnect as social conventions, values and manners are trampled when technologies intrude into our living, learning and working spaces (Plucker, 2021). Technology integration is often done by acquisition rather than design. Critical to future creative expression is creativity and innovation expressed with technology. If we seek to provide social context for groups or communities to promote their well-being, then we must create environments where they have tools and resources to explore, innovate and experiment (Plucker, 2021). Innovation has to do with coping with future events and challenges in an adequate and sustainable manner. Predicting the future; to be prepared for the unexpected, to protect oneself from possible future dangers, to make use of the unforeseen, to shape a new and unpredictable world, society, market, etc; is important for individuals and organizations. Complex knowledge processes can be found to be at the root of every innovation. This process leads to “new” insights being the foundation for a particular innovation, for a new product, service, business model, social innovation, cultural development, scientific model, etc. In terms of predicting the future shared environment, the kitchen is an important area of the home and the public space that serves many purposes both functional and social (Maguire et al., 2014), this thesis takes a look at the future kitchen design in a sustainable manner. The future kitchen is not only a place for cooking but also a place for people to express emotions, entertain and share their lives. Hence, the essence of the design of future kitchens should be based on “human life” and “human concerns” (Lau et al., 2006). As a result, it is particularly important to explore users’ ideas and expectations regarding the future kitchen.

1.2 Background

As humans, our essential nature is to be social, and our interactions with others shape our identity. The state of being alone and disconnected from others poses a threat to our overall mental and emotional health, while the act of separating ourselves from society gradually diminishes our ability to understand and accept people (Juul, 2023). An alternative social kitchen in this study refers to a communal space where numerous individuals engage in common activities such as gardening, baking, cooking, and using shared appliances to prepare meals together. The kitchen is intricately connected to individuals' nutritious diets and lives. According to (Maguire et al., 2014), the kitchen has a societal purpose and fulfils many functions. The service should prioritise comfort and functionality, while enhancing the user experience by enabling individuals to express their feelings and share their own experiences.

In Alternative Food Networks (AFN) literature, ‘alternative’ has been widely criticised for being constructed in opposition to the ‘conventional’, creating a dualism in which ‘alternative’ (associated with ‘the local’) becomes the site of social embeddedness, economic diversity, and close and meaningful relationships. Thus, ‘conventional’ (related with ‘global’) are represented by businesses, politics, markets, homogenization, and distant ties (Lundström, 2023; Misleh, 2022; Wilson, 2013). Qualifying alternatives with quality, embeddedness, and

localness has obscured the conceptual meaning of alterity. This dualism has also portrayed the alternative as a single social phenomenon rather than the dynamics of constructing alternatives with multiple trajectories (Misleh, 2022). The concept of an alternative shared kitchen in this thesis is an approach to avoiding one-sided representation and recognising a process where different outcomes may emerge.

According to (Çetin Er & Özcan, 2022), the direct and indirect effects of technology have caused both big and small spatial alterations in kitchen areas at various stages in the industrial chronology. Today's kitchens are indirectly impacted by shifting consumption patterns, such as the rise of co-living spaces and online meal delivery (Borsellino et al., 2020; Shin et al., 2019). It is anticipated that kitchen spaces will continue to need adjustment in the near future due to the quick renewal of kitchen equipment and the corresponding changes in consumption patterns (Borsellino et al., 2020). The prevalent user research approaches are linked to the communal kitchen's incomplete transformation that prevents social engagement. Even though user research and observation-based techniques like ethnography are accessible (McIntyre and Harrison, 2017, Katunsky et al. 2020). Using speculative design approaches, one may develop a user-centered knowledge of potential changes to kitchen spaces that might improve people's health and well-being in a more sustainable way. Being creative involves more than just coming up with fresh ideas; the solutions must also be put to the test in the actual world. Finding the ideal balance between creativity and conformity is necessary. According to Kurtzberg (2005) creativity is essentially a thinking issue that may be aided by an efficient problem-solving process, particularly when it occurs in groups. Prototyping speculative environments allows designers to explore future possibilities and elicit fresh perspectives.

The UN Global Goals serve as the foundation for this thesis, with an emphasis on two in particular: Goal 11 'sustainable cities and local communities' and Goal 3, which is focused on 'Health and well-being'. The study's findings may provide fresh perspectives that experts might use to start conversations with individuals in order to preserve or restore harmony in communal life.

1.3 Area of Research and Aim

Literature on social kitchen usually discusses the political aspect of the kitchen and highlight the lack of sustainable alternatives to community kitchen (Davies et al., 2019; Lundström, 2023; Maeckelbergh, 2017; McShane & Coffey, 2022). They highlight the power and social barriers, however there is a lack in addressing, understanding peoples' motivations and latent needs that could encourage shared food spaces. Moreover, their study did not take a systemic perspective into other areas that could give insights into people's need. Other literature on the kitchen space focused on the adaptive capacity of the existing building spatial layouts, technologies and circularity, however they did not examine how their interventions might contribute to the social wellbeing of the people by speculative design (Kuoppamäki et al., 2023; McShane & Coffey, 2022; Ollár et al., 2022; Özçelik & Sevinç Kayihan, 2022).

Literature on public kitchens addressed the food aspects only (Prag et al., 2023); physical and psychological needs of workers(Sørensen et al., 2016); need for more organic as well as sustainable public procurement (Sorensen et al., 2016). In their search for future kitchens, although (Z. Xu et al., 2023; Zhu et al., 2022) discussed the future expectation in the kitchen and health & wellbeing, however, they were focused on Chinese college students and the younger generation. They also concentrated on the internal kitchen and described future technologies that could be integrated in the existing building layout. Their research did not address what an alternative public space kitchen could look like. Another interesting paper (Keller, 2020) investigated the value of transforming the kitchen into a sonic performative work and public site for art and social practice (Figure 1). The public kitchen in the paper is seen as a pedagogical technique that organises and responds to the local community through creative activity and an ongoing process. It operates as an autonomous event. The paper highlights the importance of feminist struggles in navigating the self-interface, where language, meaning, and subjectivity are interconnected with human differences, social experiences, and diverse perspectives.



Figure 1: Local sound performance, in Future is Female; The risk takers (Keller, 2020)

Around the globe, communities are changing as a result of unpredictable pressures and uncertainties that require them to reevaluate their constructed, ecological, economic, and social existence. It brings up important issues about how communities might be helped to prepare for change and map a course for the future that will allow them to succeed (Sohie, 2023) more effectively. In this study we will explore to understand an alternative social kitchen and stimulate a shift towards more inclusive and environmentally conscious design practices. This is in order to explore the potential of systemic, regenerative, speculative sustainable design in addressing challenges faced by shared kitchens, while understanding diverse user needs within enabling contexts. The shared social kitchen context will be public open spaces or shared environments such as housing neighbourhoods and summerhouses. The potential of speculative design is to create new entry points towards imaginaries that can engage with complex challenges of the future while positioning creativity and design as a medium to enable informed decision making.

1.4 Research Question

This thesis aims to contribute to the design and innovation management literature by exploring to understand an alternative shared kitchen that is people-driven around shared interests, promoting social well-being. The questions are presented below.

1 How can systemic, regenerative, and speculative design be integrated to address the challenges in shared kitchen environments?
2 What are the needs and preferences of diverse user groups in shared kitchen spaces

1.5 Thesis outline

This thesis consists of chapters including Introduction, Previous Research, Theoretical background, Methodology, Findings, Discussion and Conclusion.

Chapter 1: Introduction- here I explain the background and motivation for writing this thesis. I present the area of research and aim, then the thesis purpose and research questions.

Chapter 2: Previous Research - here I present literature study on kitchen spatial layout and environment, Utopian ideas for sustainable food cities

Chapter 3: Theoretical background- here I presented the innovation ecosystem to have an overview of the actors; then speculative design and enabling design is presented to propose alternative designs for social kitchen. I used systematic design, regenerative sustainable design, to problem solving.

Chapter 4: Methodology- here I provide the overview of the methodology of this thesis. I provide a reflective description of my research journey, discuss my epistemological position and the overall research design. I provide a description of studies undertaken in this thesis with detailed information about the methodological approach in each of them.

Chapter 5: Theoretical lessons learned- here I will present the empirical material of this thesis. Discussing the purpose, findings and contributions.

Chapter 6: Discussion- here I discuss the findings of my thesis and relate them to the thesis purpose. I discuss the contributions for creativity and speculative design and the limitations of my work.

Chapter 7: Conclusion- here I provide the final summary and conclusions of the thesis.

2. Previous Studies on Kitchen Environment

2.1 Kitchen Spatial Layout and Environment

The built environment has been found to influence eating habits and healthy food promotion (Rollings & Wells, 2017). The built environment impacts human activities and well-being in various ways, from city to neighbourhood, building, furniture, and atmospherics (Wang et al., 2021). However, little is known about how home cooking and eating spaces affect food behaviours. Research has largely focused on household resources, dietary choices, and food norms (Thompson et al., 2016), but not the home-built environment (Sal Moslehian et al., 2023). The evolution of kitchen design has been studied from different perspectives to understand cultural preferences, as a market-consumption junction where food supply and preparation time is scarce (Stigzelius et al., 2018), and as an important function of the home responding to new sustainability and circularity design challenges (Ollár, 2021).

Industrialization, gender roles, architectural science, and food safety have all shaped home kitchen design. Technology has made the kitchen a functioning room with many artefacts and technology, a food environment, storage, activity space, and centre hub. This could modify lifestyle and food behaviour in smaller city flats, however the impact of kitchen design on cooking and eating is unknown (Sal Moslehian et al., 2023). Designing a place enables caretakers to cook, involve others, and build relationships, for example: increased fruit and vegetable intake, decreased obesity risk, stronger family functioning, esteem, well-being, and social participation in food preparation. This context recognises household kitchens and dining rooms for promoting and encouraging healthy eating (Dunn et al., 2023) in (Figure 2).



Figure 2: How to account for the table space, Not able to sit down as family (Dunn et al., 2023)

Sal Moslehian et al. (2023) examined kitchen design, food behaviours, and the implications of a poorly designed kitchen. They noted that size and layout, connection to other spaces, fixture and fittings (including spatial ergonomics, materiality, and texture), and indoor environmental qualities (lighting, ventilation, visual access, and window views) affect cooking/eating experiences. Despite reviewing several research, there is inadequate practical information and data to influence design choices. Mihalache et al. (2021, 2022) examined the kitchen work triangle (sink-stove-refrigerator) to see how kitchen configurations affect food

safety practices (sink-countertop-stove). Their data contribute to understanding how kitchen layouts and can be used as a starting point for future research on food safety-oriented arrangements instead of ergonomics-focused designs, food safety risk assessments; for explaining specific measures that can be implemented to improve food handling and hygiene in homes, and sociological research on consumer behaviour during cooking. X. Xu et al. (2023) measured domestic kitchen environmental characteristics by finding four factors: cooking technique, range hood, make-up air system mode, and window. Their results show that temperature, sound pressure, and fine particulate matter concentration affect the kitchen environment, allowing for improvement of comfort and health in kitchen design and renovation.

Open Plan Kitchen: In the beginning of the 21st century, the open plan dwelling – which is the connection between the living space and the kitchen, became a popular form of housing in Sweden. Concepts associated with the open plan include power, class, gender, lifestyle, identity, status and consumption. Below Maja Willén describes three perspectives of open - plan dwelling (Willén, 2012)**.**

The open plan housing as a catalyst for social life: The Swedish concept of storstuga, which can be interpreted as a big hut, serves as a metaphor for open plan living and thus validates the contemporary way of life. The theme reflects the idea that individuals are expected to lead a vibrant social existence, welcoming family and friends for dinner, offering them a glass of wine at the cooking island while the host or hostess is finalising the meal's last touches. This theme also encompasses the concept that open plan architecture consistently facilitates communication and sustains ongoing conversations by ensuring continuous contact with others in the shared space. Furthermore, it suggests that it is inadvisable to be in a home with an open plan design without any company (Willén, 2012).

The Kitchen as the heart of the home: The kitchen is a constant feature in the open plan residence and serves as the natural focal point for social gatherings within the household. Due to its central position in the open space, the kitchen design has also become a symbol of social status and thus a space for showcasing (kitchen equipment, visible cooking of the appropriate food, meals enjoyed around the kitchen table with the entire family). However, the kitchen serves not only as a physical space within a home, but also symbolises the essential function of food preparation and represents an idealised notion of a traditional lifestyle, reminiscent of 19th century cottages with their central fireplaces (Willén, 2012).

The open plan housing as a stage: The concept of the open plan home as a space for self-expression and the development of one's identity. The large windows and central open area transform the open-concept house into a stage. It serves as a setting for a life of luxury, both in a literal and symbolic sense. Visibility empowers individuals to control both the physical environment and the individuals present in the open plan area. The theme highlights a conflict between the aspiration for a large, light-filled home and the need to establish a private area where social control is possible (Willén, 2012).

Architectural phenomenology is the study of the ways in which humans perceive and interact with the built environment. Phenomenology focuses on the holistic depth of participants' or researchers' experiences, drawing generalisations about the fundamental nature of these experiences, which can be both subjective and intersubjective. Research in existential phenomenology focuses on the experiences of particular people or groups in real-world settings. The researcher must remain open to the guidance and perspectives of individual accounts in order to thoughtfully analyse and consider them, which ultimately leads to the emergence of meaningful collective themes. A post-phenomenology approach considers the multidimensionality, multi-stability, and diverse perspectives of phenomena within the context of intersubjective and subjective paradigms, with a particular focus on critical or feminist perspectives (Groat & Wang, 2013). The phenomenological aspects of the kitchen can be described based on the three-dimensional method of Fairclough (1992) and Willén, (2012); text (the objects of linguistic analysis); discursive practice (the production, distribution and consumption of texts); and social practice (Ideology and power are created at this level, which has consequences for how life is lived, and society is structured).

2.2 Roots of participatory design in the study of kitchens

The emergence of process-based and user-centered design in Scandinavian countries is usually traced to the late 1960s and early 1970s. The Home Research Institute (Hemmens Forskningsinstitut, HFI), which was established in 1944 by the collective organisations of Swedish housewives and home economics teachers, developed practices for studying work in the home. This is particularly relevant when examining the origins of the user-centered Swedish design methodology. During the establishment of the Swedish welfare state, the concept of home played a significant role as a metaphor and as an area for reform and rethinking. The kitchen became a central focus in home renovation, as it was systematically and scientifically studied to improve building standards, optimise the design of kitchen utensils and equipment, and determine the most efficient ways of working, acting, and moving within the kitchen. For example, studies were conducted by Brita Åkerman (1937,1941) and Jöran Curman (1944), in controlled laboratory-like environment and in actual homes, quantitative methodologies were used to gain insights into aspects of everyday practices (Göransdotter & Redström, 2018).

The user-centered design methods employed focused on ensuring that the design process and its outcomes are pertinent and significant to the target user groups. Efficiency, in this context, aimed at enhancing the professionalism of domestic work. The lack of emphasis on HFI in design history may be attributed to its perception as a space for evaluating designed objects rather than as a space where design actually occurs. The social design project that HFI participated in focused on effecting change through the modification of seemingly small and unimportant everyday objects. Nevertheless, these changes were conceived with the intention of encouraging distinct behaviours that could result in a transformation of power dynamics within society. Consequently, this could potentially foster a new perspective on the possibilities of societal and civilizational constructs. Interestingly, the concept of "design

research" did not initially emerge from the design field, but rather from a wide-ranging political and interdisciplinary effort aimed at promoting social change and societal development, driven by a feminist agenda (Göransdotter & Redström, 2018).

Scandinavian user-centered and participatory design: The new design methods proposed in the 1960s and 1970s sought to make visible and understandable design practices that had previously been considered individual, experientially based, and difficult to articulate. The focus of design work shifted from product to process. User-centered design entails using various methods and tools to understand different aspects of people's behaviours, actions, experiences, hopes, and dreams. Methods developed in participatory design contexts question the distribution of responsibility and agency between "designer" and "user." They aim to propose different ways to support, enhance, and collaborative and participatory practices in which designers and non-designers come together in different ways in design situations. Methods and tools are valued in user-centered and participatory design because they help with design development, experience sharing, analysis and reflection, and decision-making. Various methods such as prototypes, mock-ups, scenarios, design games, and toolkits are used to bring people together (Göransdotter, 2020).

Histories of methods, rather than things, can help to highlight why certain ways of designing have emerged, as well as contribute to critical discussions about why, what, and how different ways of current design make or unmake possible futures. The programme of transitional design histories is formulated on the notion that if design aims to develop new approaches and methods for responding to issues of complexity and sustainability, the historicity of design must be made present (Göransdotter, 2020).

2.3 Utopian Ideas for Sustainable Food Cities

For a considerable amount of time, philosophers, architects, and urbanists have challenged the notion of this post-colonial metropolis, either by reinventing the current urban environment or by developing their own interpretations of a new rural (Davis, 2016). This aims to provide a more cost-effective and sustainable option for the construction and use of the different city elements. The primary goal of their research (Jin & Zhu, 2023) is to propose a new interpretation of a self-sufficient city in response to the increasingly worsening food crisis. Frank Lloyd Wright's Broadacre City (1935), Kisho Kurokawa's Agricultural City (1960), and MVRDV's Pig City (2001) are three iconic urban utopian designs that were used in the (Jin & Zhu, 2023) project's selection and investigation of cutting-edge agricultural urbanism (Figure 3). According to (Waldheim, 2013), Wright proposed the ideas of a democratic city in which every citizen had an abundance of housing options, a variety of career options, and access to farms and gardens. The implication is that we are heading towards a future in which land and resources are becoming more costly and limited. Kurokawa's main goal in designing the Agricultural City was to reduce the likelihood of future floods by raising the infrastructure grid above the earth and freeing more acreage for future farming and flooding. The implication is that there is uncertainty about the community's connection with the land and its future plans for farming as the primary source of food. In order to construct a city for pigs, MVRDV's Pig metropolis project focused its

research and design development efforts on two important factors: density and consumption. In order to lessen the physical footprint of animal farms, it integrates all the production processes related to pig raising, processing, and transportation to concentrated high-density field towers and community services.



Figure 3: Broadacre city masterplan; Agricultural City; Pig City (Jin & Zhu 2023)

Through the perspective of utopian thinking, Jin & Zhu (2023) project aims to visualise a new vision of rural city development, one in which diverse land use plans and new population patterns are designed to foster flexibility and diversity. Since food is a natural byproduct of fertile soil, the project seeks to internalise food distribution and logistics, create new connections and an ecosystem between natural and artificial elements, and promote sustainable living by introducing new ways to use, own, and inhabit the land.

3. Theoretical Background

3.1 Innovation Ecosystem

According to Vargo & Lusch (2016), a service ecosystem is characterised as a self-adjusting system of resource integrating entities that is generally self-contained and linked by mutual creating value through service exchange. These institutions—humanly created laws, customs, and beliefs that both permit and restrict behaviour and give social interactions a predictable, purposeful quality—are the focus of Service Dominant Logic (SDL). The key to comprehending the composition and operation of service ecosystems lies in the institutionalisation process and its role. Under time and cognitive limitations, institutions allow actors to reach ever-increasing levels of value co-creation and service exchange. This is mostly because players that share institutions create a network effect with growing profits. To provide a more comprehensive and realistic view of markets and marketing, it is essential to acknowledge the crucial role that institutions and institutional arrangements play, as well as the heuristics that follow that encourage actors in a changing service ecosystem to behave cooperatively Vargo & Lusch (2016). Studies by Lindhult et al. (2018) show that virtual and systemic-based value logics can be utilised to supplement the product- and service-based logic that is the emphasis of (SDL), hence increasing the domains in which service innovation can generate value. One possible extension of SDL's value in context emphasis is to take service innovation into account within a variety of contexts. Businesses must create goods and services that are valuable and useful to consumers, suppliers, and other value network participants. A business model development can be used as a boundary object and pedagogic tool to help establish a common understanding and get all actors on board on a journey towards greater mutual benefits, based on the actors' deep grasp of value in context. Micro-institutions have a crucial role in promoting, guiding, and limiting innovation. Strong cultural and intellectual leadership must be developed in order to foster innovation through various value logics. Enabling spaces and settings must also be created.

3.2 Responsible Innovation

The increasing usage of complex technology in society affects the uncertainty encountered by individuals who assume responsibility for their actions (Grinbaum & Groves, 2013). We must plan ahead to innovate. Relying on knowledge for foresight and focusing on the repercussions is insufficient. Instead of worrying about the future and non-humans, we should be inspired to innovate responsibly. Approaches to responsible innovation include Passion and reason: we must acknowledge that our circumstance is not ethically unusual and learn from previous moral thinking about ethical ideas that may be applied to our present situation. non-consequential individual responsibility: innovators' future-oriented responsibilities, based on vulnerability, mediate between caregivers and innovators; collective political responsibility: historical continuity from past to future. The innovator must consider the social and political implications of their goals and the measures required to achieve them. A virtuous innovator must act to fulfil lifetime obligation (Grinbaum & Groves, 2013).

Systemically using Responsible Research and Innovation (RRI) methods transforms futures (Richard. Owen, 2015). RRI is a democratization process connecting science to the values of people by participatory processes. What futures do we want science in innovation to generate and how can we do so inclusively? We must grasp the ethical, social, environmental, and political aspects of science, technology, and innovation today. It acknowledges uncertainty, ignorance, and ambiguity (Richard. Owen, 2015). Doing good is normative: i.e,. What innovation is right? What is the right influence of science and innovation? What values should guide innovation research as a future-making process? We must commit to sustainability, privacy, ethics, and not forcing our ideals on the world (Richard. Owen, 2015) are some of the issues that RRI focused on. Wickson & Carew (2014) developed seven linked and interwoven RRI quality criteria to concretize quality concepts. They suggested evaluating RRI by its social relevance and solution-oriented, sustainability-centered and future-scanning, diverse and deliberative, reflexive and responsive, rigorous and robust, innovative and elegant, honest and accountable. This will spark discussion among innovators, researchers, project leaders, and others. For instance, the New European Bauhaus movement promotes a sustainability culture among actors in the meta-innovation ecosystem (Rosado-García et al., 2021), transforming our ways of accounting, interacting with spaces, living and working, and designing our built environment. Stahl et al. (2021) highlighted that data protection and research ethics compliance are covered under the GDPR. It protects individuals' rights and freedoms regarding personal data, especially research data, while processing it. Consent and participant protection govern human research ethics. RRI must be part of the research and innovation process, not an external requirement/complement, to succeed.

3.3 Spaces enabling and sustaining innovations

Complex knowledge processes underpin all innovations. An enabling context (not controlling or managing) is needed to foster knowledge and innovation. Enabling Spaces (Peschl & Fundneider, 2012) involves creating and designing environments to facilitate cognitive processes of creating new knowledge by establishing a smooth, stimulating, and fertile interaction between these domains and the participating cognitive system(s). To understand sustainable and successful innovation and how to facilitate it, we must study these cognitive processes. We also need to study how they interact with the environment and generate (radically) new knowledge and innovation. Innovation is always the result of a complex network of actors, dynamics, and constraints. Permanently interacting sources of knowledge and innovation include: 1. Object of innovation: to create sustainable and game-changing innovations, you must understand its core. 2. Users/Market: Innovations must resonate with users and offer something new in this domain. 3. Society: all innovations are embedded in a society with cultural constraints, values, etc. 4. Technology: technology is crucial to most innovations because it provides the “mechanism” that makes the system work (Arthur, 2007). 5. Organisations are the “structural container” for most industrial innovations. Organisations usually provide stability (“retention”) in highly dynamic and volatile innovation dynamics (Peschl & Fundneider, 2014).

Enabling requires giving up control, rigidity and making. (ii) Enabling means providing constraints or a framework to support new knowledge creation. Similar to Dorst (2003)'s design problem of reframing before solving. This implies an autonomous, dynamic knowledge field that changes its basic parameters over time. This is also known as the self-transcending knowledge by (Scharmer, 2007). Emergent Innovation, as described by (Peschl & Fundneider, 2014) is based on the innovation paradigm and a systematic innovation process that generates new knowledge. The Enabling Spaces framework acts as a container that holds innovation processes and activities. The multi-dimensional Enabling Space integrates architectural/physical, social, cognitive, technological, epistemological, cultural, intellectual, emotional, and other factors to support innovation. Enabling Spaces aims to integrate these aspects into a single, interdisciplinary design. Every routine and social interaction in a concrete enabling space requires trust (Peschl & Fundneider, 2012).

3.4 Creative problem-solving

Design thinking is gaining interest in the management profession as an effective way of working that can foster and encourage innovation (Beckman & Barry, 2007; Boland et al., 2008). According to (Goller & Bessant, 2017), innovation is the fundamental renewal process that occurs inside an organisation and produces new goods and services that benefit the user as well as the business. A multifaceted concept, design thinking has been interpreted differently by designers, innovators, and managers of early-stage innovation activities. For example, Design thinking, according to (Cross, 2006) is "the core creative process for any designer." Dorst (2011) sees design thinking as a collection of many cognitive processes that include abductive, deductive, and inductive reasoning. Statistics merely shows the “What” of data; but finding the “Why” requires creativity. Creativity gives organisations a competitive edge by looking forward. A competent leader should create enabling environments for idea generation and selection to emerge from colleagues or staff (Madden, 2017). Successful new product development (NPD) efforts focus on a deep understanding of customer needs, require cross-functional involvement, have good sponsorship, have a clear and compelling strategy, and manage creativity (Isaksen & Tidd, 2006). Vision, external communication, support for innovation, task orientation, internal communication, cohesiveness, and goal interdependence were shown to be strongly and generalizable related to innovation (Hülsheger et al., 2009). According to Hargadon & Bechky (2006), collaborative creativity occurs when social interactions lead to novel interpretations and distant analogies that people alone could not have developed. Their field research indicated four interrelated actions that drive collective creativity: assistance seeking, help giving, reflective reframing, and reinforcing. Formal databases may be unproductive because asocial behaviours hinder deliberate reframing and evaluating old concepts in new circumstances. The interactions that turn people into creative collectives should be reinforced by leaders. Organisations that want to survive and flourish must address the innovation problem systemically (Isaksen & Tidd, 2006). Taking a systemic approach requires intentional, deliberate, and disciplined efforts that include all key system elements -people, processes, product, and place.

3.5 Systemic Sustainable Design

In a systemic approach, people's concerns are crucial to change implementation, along with the situation, environment, and culture. It also involves examining other ways to achieve the desired outcome. In volatile and fast changing times, multiple change management strategies are essential to survive. Organisations must regularly update their goods and processes. Successful organisations have careful idea management, a creative climate, and an inclusive leadership (Isaksen & Tidd, 2006). Design abduction involves creative, logical, critical approaches to frame the challenge and solution. Design techniques must simplify and overcome designers' cognitive constraints like fixations to cope with information overload. Complexity requires a major transition from problem-solving-based design paradigms to complexity theory and systems thinking (Ball & Helbing, 2012). Creating resilience of a continuing adaptive design process rather than seeking a solution can produce newness, change, and relative stability. To improve the system, high-quality interventions are needed. This is an explorative, reflexive, practical approach to designing. Designers should abandon problem-solving and embrace the complexity of the design issue to create new, more exploratory design processes (Dorst, 2019).

Design council highlights six principles for systemic design; people and planet centered: focusing on shared benefits of all living things; zooming in and out: from present/ personal to future/ wider system; testing and growing ideas: making things to emerge; inclusive and welcoming difference: creating safe and shared space and language to multiple perspectives; collaborating and connecting: seeing a project in a wider movement for change; circular and regenerative: focusing on existing assets and how we can re-use, nurture and grow. Beyond the innovation process and taking a systemic approach to innovate involves ISO56000 perspective (International Organisation for Standardization). An organisation can innovate more effectively and efficiently if all necessary activities and other interrelated or interacting elements are managed as a system. Innovation management system framework (Karlsson & Magnusson, 2019) contains elements that are categorized under eight themes of; context, direction, leadership, culture, processes, structures, support and resources, and evaluation. This is to ensure a sustained renewal capabilities and continuous creation and realization of value for the organization and its stakeholders.

3.6 Regenerative Sustainable Design

Regeneration is an important process of biological systems and refers to the capacity to bring into existence again. The strains in ecological, social, and psychological realms reflect a broken paradigm that cannot solve present challenges. Ecoism (ecological denial) is embedded in systems and worldviews. The existing priority and value order is unsuitable. Eco-social crises will worsen unless we adjust. Designers should materialise ideas to create objects that are meaningful (Boehnert, 2018). In the welfare state, sustainability and resilience become robust with a flexible generation system is implemented (Forsberg & de Souza, 2021). Giusti & Samuelsson (2020) mentioned the regenerative compatibility between healthy ecosystems, environmental attitudes, and restorative experiences in their research.

Regenerative compatibility could help eliminate artificial dichotomies between people and environment, healthy humans and a healthy biosphere, and natural and human habitats. Health, environmental education, and urban ecology should be addressed jointly to build sustainable cities. Such techniques are key to moving from a static, compartmentalised perspective of sustainability to one that is comprehensive, dynamic, and regenerative.

Regenerative business models concentrate on social-ecological systems and contribute to sustainable and circular business model literature (Berkes et al., 2000). They propose a new goal and framing of planetary health and societal wellbeing and a motivating narrative that rethinks the relationship between humans and nature, not just net zero or triple bottom line. Regenerative business models also emphasise personal purpose, leadership, spirituality, environment and animal rights, and indigenous cultures' role in reconnecting people and nature. In their model (Figure 4), regenerative, sustainable, and circular business models share design principles: Design for the regenerative biosphere combines circular and regenerative themes. Sustainable and circular link via circular technology, and sustainable and regenerative business models through sustainable persons (Konietzko et al., 2023). The model helps to advance organizational efforts to evolve humanity towards planetary health and societal wellbeing.

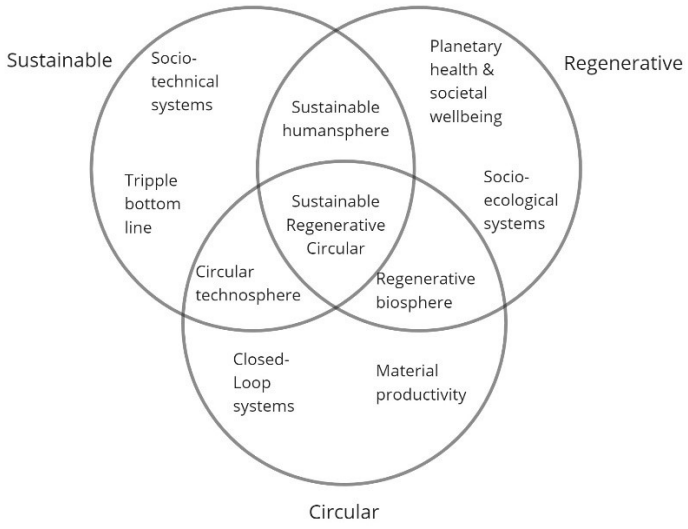


Figure 4: Regenerative business model (reproduced from Konietzko et al., 2023)

By leveraging existing successful ecovillage development and technologies such as ecological design and planning, regenerative development, and regenerative design, these tried-and-true techniques and methodologies could be applied to a greater extent via regenerative community development, regenerative landscape development, and regenerative community development (Gibbons, 2020).

3.7 Speculative Sustainable Design

Norman Henchey (1978) defined desirable, plausible, and feasible futures to explain future study methodologies. Dunne and Raby's (2001) worldview of critical design contribute to the notion of speculative and conceptual design, as ways to probe and provoke the domains of values, worldviews and presenting alternatives in problem solving. 'Futuring' design, according to Tony Fry (2009), sets new paths for design development and provides a more sustainable future. Design study on attitudes, beliefs, and behaviour asks what if things were different. Presenting alternatives makes various now more present (Redström, 2018).

Speculation can be hampered by the complicated, relational, material, and unknown future (Koro & Tanggaard, 2022). It is more concerned with accepting that all things are equally real and that research matter's agency is feasible and plausible beyond human knowledge, language, and awareness than with how materiality and research matter may communicate back or have human agency. From this viewpoint, our world is continually changing and uncertain. Furthermore, speculation methods play, experiment, shift, generate issues, and inquire about possibilities and "what ifs." Speculative research methods wonder but seldom verify, solve, or claim to comprehend others. Instead, they may care, connect, and make "educated guesses" and possible scenarios using speculation's inquisitive, creative, and visionary skills (Somekh, 2007).

Different creative research methodological possibilities require significant transformations in thinking about how, where, when, and why methodologies and study approaches work and for whom (Koro & Tanggaard, 2022). Speculative methods that emphasise ecological and social thinking may aid future creativity study. For example, Titchen and McCormack (2010) used critical creativity in their action study. The research method was a dance of dimensions, knowledge, discourses, and creative imaginations. Creative arts media, walking and being in nature, artefacts of artistic expressions, physical and metaphorical dance, and creative writing were also employed to promote transformative and humanistic critical creativity. Another model of future desirability in temporal work, Alimadadi et al. (2022) highlighted how agents generate, connect, and traverse views on desirable and unpleasant futures to form a route of action. Angheloiu et al. (2020) shows that dominating future imaginaries include breakthrough technologies solving sustainability problems. Sustainability requires changes in 'interiority' or 'inner dimensions' (Wamsler & Brink, 2018), the 'subjective domains inside the person pertaining to people's mindsets, worldviews, attitudes, values and emotions. They noted difficulties in addressing Western imaginaries' homogeneity and late modernity's hyper individualism. Fostering pluralistic future imaginaries is crucial.

Discussion/Summary of the Theoretical Framework

The theoretical framework is designed to understand the shared kitchen environment through the lens of (1) exploiting the value creation process of the people involved in the context of the organisation / neighbourhood -Innovation Ecosystem. (2) It also seeks to understand the strategic routines and preconditions that will be sustainable for continuous success by understanding Responsible Research innovation. (3) knowledge creation processes and innovation processes are relevant tactics for continuous innovation during creative problem solving, speculative design and spaces enabling innovation. Through committed leadership and innovation strategy of ISO 56002, the shared kitchen environment incorporates systemic sustainable designs and regenerative sustainable designs.

4 Method

4.1 Research Design

I have designed and supervised the construction of wet areas such as the kitchen and bathrooms while working as an architect and project manager. My previous office renovation project includes the use of specified imported joinery and finishes e.g Thrislington UK for toilet cubicles; Gamadecor and Porcenalosa for kitchen joinery, toilet sanitary and tiles; Saporiti for joinery; and Lualdi for glass/wall panels and doors. Kitchen appliances such as Bosch, Miele and Electrolux were also used depending on the client's budget.

As an architect, I am interested in how architects and designers can be activists towards social change and how architecture can be used to positively change behaviour and society. According to (Boehnert, 2018) design activism works to confront capitalist developments and create post-capitalist alternatives. The kitchen for me serves as an artefact to delve into many latent issues in healthcare and welfare. As mentioned in my thesis literature, there is limited understanding of people's needs in the kitchen from a systemic perspective and how the kitchen design influences their lifestyle. The kitchen serves as a tool of thought that can evoke emotional involvement from diverse voices, sharing their personal lifeworld experiences. Following learnings from Frank Gehry's design practice (Boland et al., 2008), I adopt a design attitude and shift from stereotypical behaviours (of maintaining status quo) to creative problem solving and seek the betterment of shared kitchen environments. My codesign approach in making (E. B. N. Sanders & Stappers, 2014) in the fuzzy front end of the design process will be to work with other designers and non-designers, using creativity to make sense of alternative social kitchens that are more inclusive and sustainable. Interpretation and explanation are interdependent in hermeneutics (McFadden & Ricoeur, 1978). Explanations can help researchers go beyond the text's conditions and meanings to find hidden meanings. An open, pliable, reflective, and problem-solving hermeneutical attitude is needed (Palmér et al., 2023). Is there new data? Does anything contradict my knowledge? With regards to kitchen design, I aim to challenge "kitchenly" norms. There are assumptions that the kitchen is a woman's domain; that the current modular kitchen is perfect for everyone, or that a social/community kitchen fails to reflect a prosperous and innovative community. Critical realism helps us argue that science should generalise and reveal social structure causal mechanisms. Abduction interprets and recontextualizes individual phenomena (Danermark, 2002) to help understand the use of the kitchen in a new way. In principle, power is everywhere. Power is the result of the dominant class's strategy, not a privilege. Power produces knowledge and is based on knowledge. Hermeneutics will be employed (Alvesson & Sköldberg, 2018) to comprehend the social welfare of the kitchen home care and community. The focus of this research is to explore an alternative social kitchen and stimulate a shift towards more inclusive and environmentally conscious design practices. This is in order to explore the potential of systemic, regenerative, speculative sustainable design in addressing challenges faced by shared kitchens, while understanding diverse user needs within enabling contexts. The approach is to alternate between design and learning practice as shown in (Figure 5).

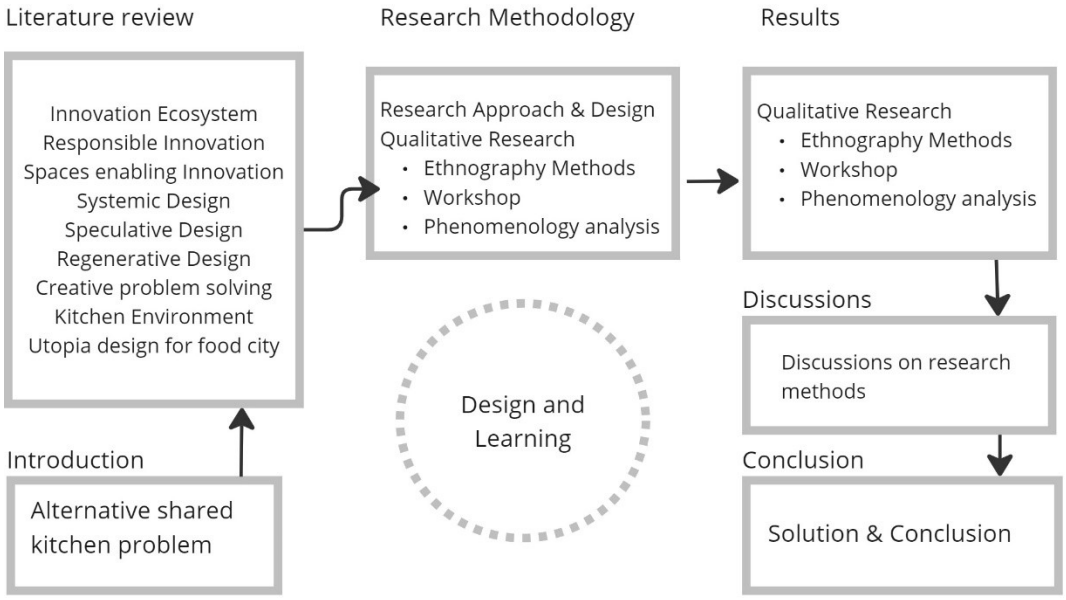


Figure 5: Research project

Given the explorative nature of this thesis is to understand an alternative social kitchen to promote social well-being, a qualitative approach is appropriate and flexible to allow the consideration of different aspects of a phenomenon (Kothari, 2004) as seen in (Figure 6). Qualitative methods were chosen since it uses interpretive frameworks in addressing the meaning ascribed to a social/human problem. The goal of qualitative research is to understand events in their specific context and how they are interpreted within it (Table 1). It aims for a comprehensive knowledge by attempting to comprehend the nature of the context, what it means for individuals to be in that setting, and what their meanings are (Merriam & Tisdell, 2016). I will use multiple sources for data collection, such as interviews, observations, documents and artefacts (Creswell, 1998). Under qualitative method is Ethnography research, which I will use because it focuses on the manner in which people interact and collaborate in observable and regular ways. It requires immersion in the culture being studied which includes organisational cultures, by writing accounts of people's subjective experiences, in thematic narrative, over a considerable period of time. Data is collected through participants observations, interviews, artefacts, documents of a single culture-sharing group. Information will be recorded using field notes, interviews, observational protocols (Fisher & Buglear, 2010). Data will also be collected in design workshops (Martin & Hanington, 2012), where I will facilitate in designing conceptual solutions for social kitchens. For validity, I will provide enough information about the context to arrive at the conclusion. My reflexive approach as a critical realist will ensure that all perspectives and interpretations are covered and factors from different actors in the kitchen service (Fisher & Buglear, 2010). Phenomenological method will be used to analyse the data for significant statements, meaning, textual and structural description, and description of the essence of care needs (Creswell, 1998). The study of phenomenology focuses on how individuals perceive their conscious lives. It provides us with a means of pre-reflectively seeing the world. This kind of study is predicated on the assumption that shared experiences

have a fundamental quality. The essence is the fundamental meaning that all parties understand from a shared experience of an event. It is often understood to be the study of people's subjective daily experiences. It involves a process of breaking down data into codes and meanings include writing down important quotes, organising the data into units of meaning, and creating summaries that capture the core of the events (Creswell, 1998).

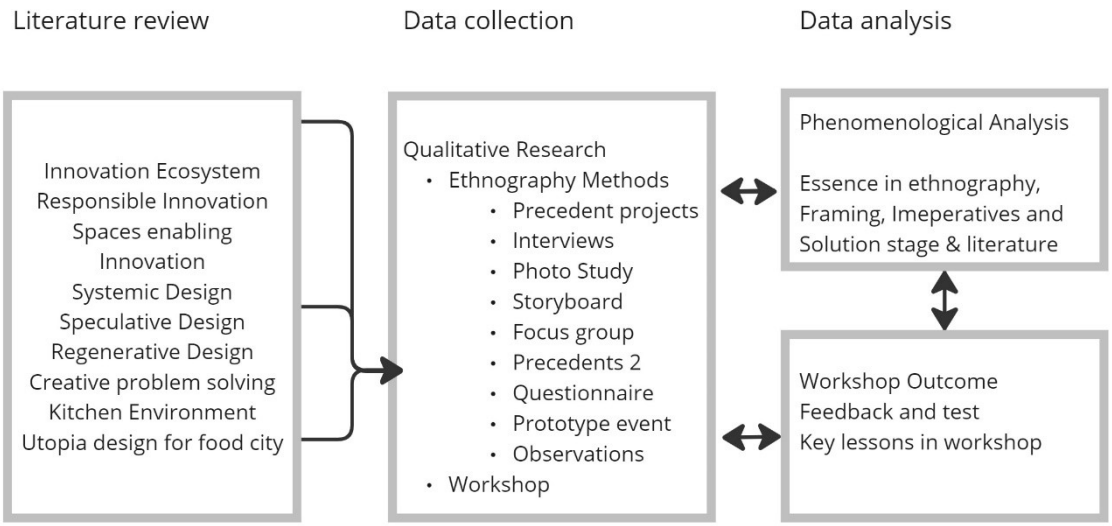


Figure 6: Research design

This thesis uses a hermeneutic reflexive interpretation perspective for its ontological, epistemological, and methodological assumptions (Alvesson & Deetz, 2011). Organisational reality is not static and 'out there'. The researcher and others create their own 'reality' at this time and place (Pärl, 2013). Understanding a human's existential issues and lived experiences in a specific life situation requires ontological and epistemological foundations and reflections on what it means to be human and how human understanding is possible. According to Gadamer (1978), prejudices shape our perceptions and understanding. Research based on interpretations makes it difficult to see something new or different from the researcher's pre-understanding (Palmér et al., 2022). This is in line with (M. Gibbons et al., 1994) mode 2 knowledge that stress analysis and design should be carried out in a continuous process with the field and in collaboration with participants.

	Method	**Aim of Method**
Preliminary phase	Literature Studies	Gain knowledge about the subject area
Data Collection	Ethnography Research Workshop	Gain core knowledge and meaning about users experience and lifeworld
Data Analysis	Phenomenology Analysis	Categorize the data based on research question

Table 1: Aim of Method

4.1.1 Research through Design - Design as a process.

This thesis will view design research and the design perspective as knowing and social processes. Social change, transformation, and improvement can result from research through design (Cross, 2006). It will reveal intentional, radical knowledge changes through design. Kitchen users, facility managers/caretakers, and the community will share their experiences and lives in the project. Design is nonlinear and uncertain, requiring exploration and periodic reflection, review, adjustment and trust (Olsen & Heaton, 2010). Designing requires elements of mindfulness, the thoughtfulness based on cues signalling that one's frame and theory of the situation should be altered (Olsen, 2008).

Critical approach and reflexivity in transformation design challenge power relationships and demonstrate how to envision, initiate, and sustain change. Transformative design principles include active citizens, community-scale intervention, building capacities and project partnerships, redistributing power, designing enabling platforms, enhancing imaginations and hope, and evaluating success and impact (Sangiorgi & Scott, 2014). Meaningful innovations improve economic and social conditions and company competitiveness. Shared value creation looks for ways to link social and economic progress. Ecosystems function as central platforms for actors to create or improve services. The value framework uncovers meaningful innovations and value opportunities (Den Ouden, 2012).

Below is a summary of my approach (Figure 7) and the use of literature to capture learnings to solve the overlooked or the unaddressed problem of kitchen design and how it affects people's wellbeing unknowingly. The thesis aims to understand users' needs and explore an alternative shared kitchen, promoting social, mental and physical wellbeing.

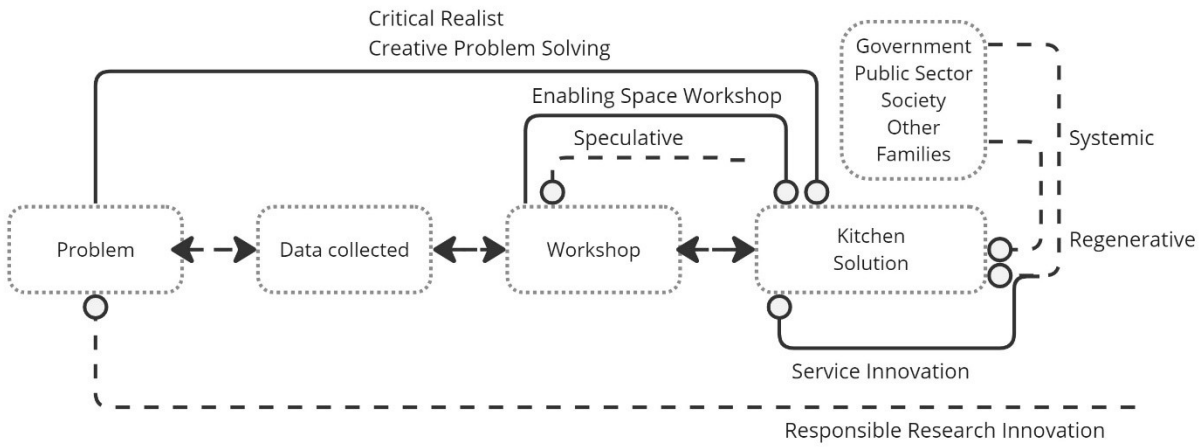


Figure 7: Research approach (Akintade, 2024)

4.2 Innovation Process

The innovation process and learning model (Figure 8) involves both analytical and synthetic components, and switches between the theoretical (abstract) and practical (concrete) domains. In this model, the process of reflection is used to convert observations or experiences in the concrete world into frameworks or insights. Frameworks are transformed into imperatives or ideas through a process of convergent thinking. Imperatives are translated into tangible solutions by implementing actions in the concrete world once more (Beckman & Barry 2007).

This model closely aligns with experiential learning theory models (C. L. Owen, 1998), which propose four learning styles that correspond to the stages of the design process: divergent learning styles are most suitable for the observation task, assimilating styles for the framing task, converging styles for creating imperatives, and accommodating styles for developing solutions. In this thesis, each of the four non-linear activities involved in the design process is explored: observation (ethnography methods), frameworks, imperatives, and solutions.

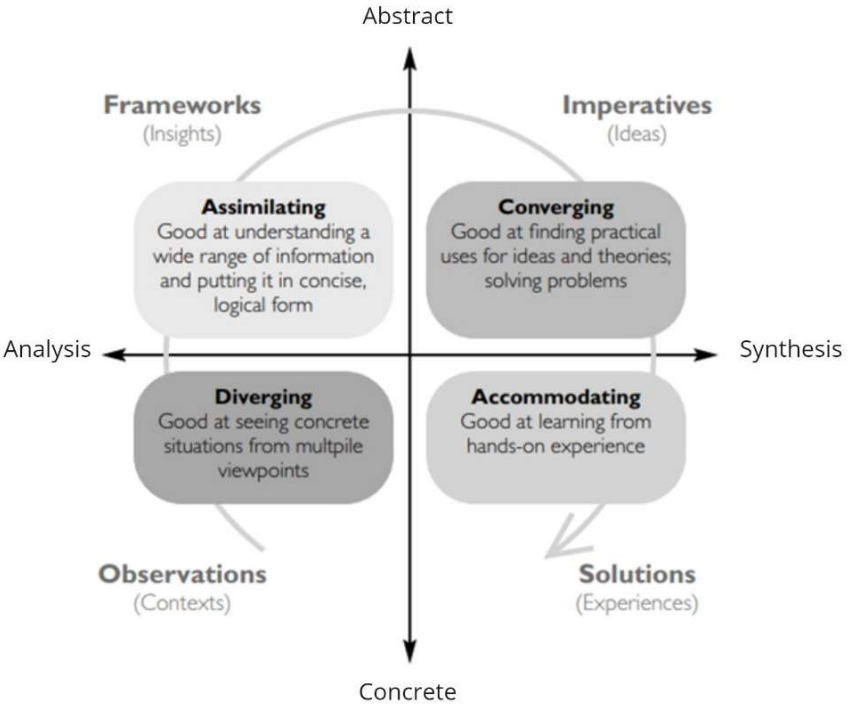


Figure 8: Innovation Process and Learning Styles (Beckman & Barry, 2007)

4.3 Ethnography Methods

Ethnography research seeks to understand people's motivations, expectations, meaning-based needs. Engaging in good observation activities such as contextual inquiry, ethnography market research, onsite observation, that provide the designer to understand how the services being studied are used and how its benefits are derived in context of use (Merriam & Tisdell, 2016). In this thesis project, for data collection and learning, I will use the four stages of the innovation process including – observation (ethnography methods), frameworks, imperatives, and solutions (Figure 9) as shown below.

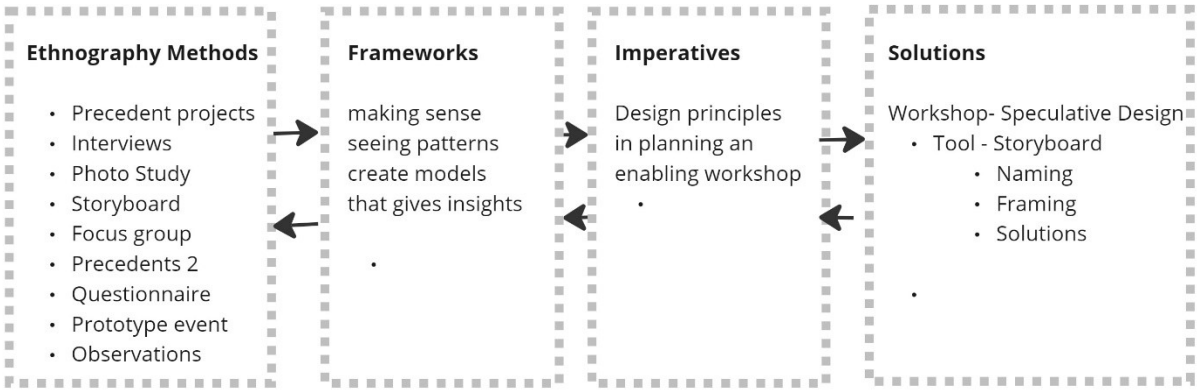


Figure 9: Innovation process and Learning styles for data collection (Akintade, 2024)

The ethnography methods conducted include: precedent projects of alternative kitchen designs, semi-structured interviews, photo studies, storyboard, focused group studies, questionnaire, prototype of a food event and observations.

4.3.1 Description of Each Ethnography Methods

Precedents are collected pictures, illustrations that can be used to visually communicate an essential description of targeted aesthetics, style, and context of the design intent. It can serve as a tangible focus for the designer for inspiring their designs (Martin & Hanington, 2012); *Application of Precedents:* to gain understanding about the context of the kitchen and get inspiration from similar ideas of a future kitchen, I searched online websites (e.g Archdaily and Dezeen) for related projects using the search string "future kitchen". The pictures chosen were those connected to sustainability and targeted at users and environmental context. It describes how the users would interact in the kitchen environment. A second iteration of the precedent was done, this time, related to kitchens in the tropical regions, and modern kitchens related to neighbourhood food networks. The precedent is used to establish previous projects on the subject matter and identify opportunity gaps to help suggest a research direction and method.

Semi-structured interviews are needed for understanding the phenomenon under study, it includes- flexible and less structured questions, usually specific data required from all

participants, though guided by a set of questions to be discussed there is no prepared wording or order of question that is predetermined (Merriam & Tisdell, 2016). *Application of Semi-structured interview:* this is to collect first-hand accounts of people's changing lifestyles, distinct needs, experiences, expectations of technology-adapted users and motivation. On March 4, 2024, two (2) people participated in a telephone interview, with a duration of 15 minutes each. They were selected because they have lived in different types of housing units and are migrants having stayed in their new home country for about two decades. One of the participants was an architect (A); the other participant is a non-designer, in a nomad profession as a groundwater modeller (C). The questions asked the two participants were: What do you like about your kitchen? What don't you like about your kitchen? What is your eating lifestyle? Further in the thesis, I wanted more professional knowledge on food wastage and how the kitchen can be improved. This led to an interview with a third interview on 15 April 2024 with a mechanical engineer (M) lasting 30 minutes. Questions such as: How can building services be improved to reduce food wastage? How can we improve socialization in neighbourhoods? All the three interviewees gave their consent for the purpose of this thesis.

Photo studies is an exploratory method for understanding the world of the users. It involves the participant in a photo-document aspect of their life and interactions, providing the researcher with visual, self-reported insights into the users' behaviour and priorities (Martin & Hanington, 2012). *Application of Photo studies:* This is to understand the world of different people using the kitchen. One (1) participant was involved in the photo study. The participant used his own camera to capture pain-points in his shared kitchen. He provided visual samples of the important things in his lifeworld to inform and inspire the design of the alternative social kitchen. This was used to capture information that can form a basis of conversation and can uncover latent needs of the participant.

Storyboarding can be used to capture important social, environmental, and technical factors that shape the context of how, where, and why people engage with services. By using rich narratives, storyboards can be used to build empathy for end-users and reframe touch points and consider alternative designs in the early phases of the design process (Martin & Hanington, 2012). *Application of Storyboarding in this Ethnography stage:* the storyboard involved one participant (1) who has artistic capabilities as an illustrator however he did not design or draw the storyboard. As we discussed online, he referred me to the film "Back to the future 2". This was used to provide a visual narrative that communicates the context of his description of what a future kitchen might look like.

Focus group studies is a group setting of chosen people, guided by the facilitator to give deep insight into situations, patterns or trends. It is an interactive discussion through which data are generated while participants share their views and listen to others (Merriam & Tisdell, 2016). It involves gauging opinions, feelings, and attitudes about the discussions on the lifestyle in the kitchen/home. Attention needs to be paid to the logic behind conclusions, stories, metaphors, analogies, as well as how participants describe their experiences, preferences and memories. *Application of focused group studies:* The focus group activity includes ten (10) participants who come from all sectors, industries and professions; and meet once a week to have lunchtime coffee together, get inspiration and talk in groups. The age range of 35 – 65

years old and Swedes. The location was a “virtual penthouse” using welo online meeting platform. My role included presenting the idea to all nine participants, facilitating, observing, and asking questions in the smaller group of three: observing and listening during final discussions.

Questionnaires are data obtained from the participants on their attitudes, views, feelings, and ideas. *Application of Questionnaire:* In this thesis, I prepared open-ended questions for in depth responses related to likes and dislikes about the kitchen; and how people’s different professions could make it easier to use the kitchen for social interactions, functionality, healthy eating and reduce food wastage.

Prototyping includes the production of artefacts for the ideation and testing process, which is important in the early stages of innovation. *Application of a food event prototype*: I included in this section a past project related to food. In Spring of 2023, during a project management course, I and two colleagues worked on the vision of a food festival prototype at Mälardalen University with the purpose of promoting collaboration and integration among Erasmus students, international students, and Swedish students of different ethnicities. Asides planning and documentation of the write-up, my role was to implement the food event. Seven (7) countries participated in cooking at the event, and we had diverse visitors.

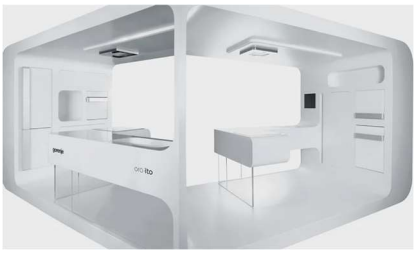
Participant observation is an intensive approach to ethnography that uses people' experiences participating in activities, contexts, and cultures to analyse circumstances and behaviour (Martin & Hanington, 2012). *Application of Participant Observation:* This includes my observation of social interactions in my student housing block and at a social event. As an international master’s student, I have a two-year experience living in two student housing apartments. Both are 11-storey, single room apartment with private bathroom, kitchen, bedroom and balcony. To understand social possibilities, I attended “Festen" by Victor Fröjd (Västerås, April 11, 2024), an innovative, genre-crossing performance with impressive dance performances and a liberating lack of prestige.

4.3.2 Findings of Each Ethnography Methods

4.3.2.1 Precedent Projects – Future Kitchen

Precedents: There were many interesting projects concerning the future kitchen that contributed to this thesis project described in (Table 2) below.

	Caesarstone and Pratt: In 2017, Following a collaboration with Pratt Institute, Industrial Design students were challenged to combine theory and creative application in their vision for the kitchen of 2050. Caesarstone and Pratt's future Kitchen is an experiment in how nature and technology may collaborate to create a self-sustaining place powered by aquaponics, hydroponics, 3D printing, water filtering, indoor farming, composting, and food storage.

	During IFA 2016 French designer Ora-ïto believes that kitchens in the future will be easy, connected, and exciting.´I believe that the functions of the kitchen will be spread out throughout the home. appliances will also provide connectivity, both among themselves and with the user–so you will be able to start cooking before you even get back home. entertainment and interactivity will become core elements of every product, making it more fun to use'.
 	Berlin- and Vienna-based Chmara. Rosinke presented four design concepts in a Milan design week show titled The Kitchen for Cooking, which was exhibited as part of the Alcova exhibition. Freed from the constraints of a typical fitted kitchen, the designs combined familiar kitchen elements in surprising ways, often with unusual shapes and playful approaches to storage. "Cooking is more than just food preparation," they said. "From culture to culture, the act of cooking varies greatly and differs from the image that is in our heads. It is a multi-sensory activity that can play a whole new role in our world that is increasingly taking place in the digital space." Made from stainless steel and plywood, its form was developed with consideration to how energy efficiency and recovery can be optimised in the kitchen, and how cooking habits have changed since the pandemic. The design includes an irregularly shaped counter, an overhead bar for hanging utensils, storage space for home fermenting and an oven on wheels.- 2021
	Sustainability at the forefront as Electrolux presents future kitchen concepts at Eurocucina. "By influencing how our products are used, we believe we can help reduce food waste, promote resource efficient food choices and nutrition, and enhance healthy and sustainable eating experiences," - June 7, 2022
	In 2015, a team of students created a concept kitchen table for IKEA, which acts as an integrated cooking hob and dining table and can suggest recipes based on ingredients you put on it. The interactive table can identify an ingredient placed on it via a camera and image-recognition technology. It will then suggest other ingredients to combine with it and can take the user through a full step-by-step recipe via instructions projected directly onto the surface of the table. The idea behind the table is to promote cooking and reduce the amount of food that gets thrown away.

	Industrial designer H. Creston Doner had a chance to build the "Kitchen of Tomorrow" for the Libbey-Owens-Ford Glass Company, he did so with his wife in mind. The resultant kitchen was profiled in a 1943 issue of Life magazine. With sliding cabinet doors, plentiful glass and working-height counters.
	Kitchen is a cooperative initiative supported by the region of Västra Götaland that seeks to provide guidance and training for work in communal food settings.
	Tele-robotic project allows online users to view and interact with a garden filled with living plants. Users can plant, water, and monitor the progress of seedlings through movements of a robotic arm – Goldberg and Santarromana 1994

Table 2: Precedents of future kitchen designs

4.3.2.2 Semi-structured Interview

As mentioned earlier, two (2) people participated in the semi structured interview on the same day (Table 3). One of the participants was an architect (A); the other participant is a non-designer (C). We discussed their lifestyle using the kitchen. They both enjoy eating their cultural dishes and staying healthy with good eating and exercise. While (C) felt his kitchen is adequate with fridge, storage, and good lighting, Participant (A) had reservations towards his kitchen.

> My kitchen is too narrow, the extractor is low, and I sometimes hit my head while cooking, the fridge is too far from the cooking area. In my previous apartment, I did not like the kitchen opening to the living area and seeing all the kitchen appliances (A).

Code	Profession	Location	Date	Mode of communication
A	Architect	UK	4 March 2024	Telephone
C	Ground water modeller	Australia	4 March 2024	Telephone
M	Mechanical engineer	UK	15 April 2024	Telephone

Table 3: Semi-structured interview

The third interview with the mechanical engineer(M) was conducted to explore if new technologies can sort and extract food waste directly from the kitchen to farms (bypassing the bin house). He highlighted that sorting food is very tricky. It is a combination of food that decomposes easily, smells and forms bacteria. Oil is also not good for soil and biological life. The waste can be used for biomass for electricity. He mentioned that a kitchen technology that saves wastage is the air fryer which can do multiple tasks. He mentioned that the size of the sink, use of a sieve, and setting the right temperature for cooking reduces wastage. Food events like during the Ramadan can connect people to socialize in schools, charity homes or public spaces.

4.3.2.3 Photo Studies

The same participant (A) in the semi-structured interview, also engaged in the photo study exercise. He used his personal camera to capture pain-points in his shared kitchen apartment (Table 4).

Participants photo studies	Notes
 	….the shared apartment kitchen is too narrow, and the fridge is far from the cooking area. No cross-ventilation. I sometimes hit my head on the extractor while cooking.

Table 4: Participant's Photo study

4.3.2.4 Storyboarding

To explore more ideas of an alternative social kitchen that considers sustainability, I engaged two (2) participants to speculate on how the kitchen would be designed in 50 years considering future lifestyles of people and relationship with food. One of the participants is a male storyboard artist and illustrator, Kro Onimole of Crasherscomics (T) and the second participant is a human resource manager (H) - female. They both gave verbal scenarios, and I created the storyboard (Table 5).

Code	Profession	Location	Date	Mode of communication
T	Illustrator- Kro Onimole	UK/Nigeria	7 March 2024	Telephone
H	HR manager	Canada	7 March 2024	Telephone

Table 5: Scenarios for Storyboarding

Idea generation and use of storyboards: The question that I asked Participant (T) was: how can we speculate with design fiction, how the kitchen will look like 50 years from now

considering new lifestyles (Appendix 2)? I also asked if his female comic character "Risi" acted in a futuristic kitchen. He mentioned that Risi cannot cook, and he referred to the film "Back to the future 2". (I didn't ask him to draw, because I knew how artists/ designers/ architects could be exploited to give a design/sketch without payment). Hence this informal route. He said in the future, "you put packages into a machine, and it prepares it into a food". Back to the future 2, is a 1989 American science fiction film written by Robert Zemeckis and Bob Gale of time travellers from 1985 to 2015. In a separate conversation with participant (H), she mentioned "what if the kitchen is invisible", and we both laughed at the ridiculousness of the idea. She iterated that what if the kitchen could be hidden away and you don't see it once you have completed cooking. I then watched the movie on my Viaplay and screenshot the scene "you put packages into a machine, and it prepares into food", the movie's kitchen scene also has a drop-down garden which was included in the storyboard that I created because it is related to food in a dynamic/flexible kitchen (Table 5).

Participant (T)'s storyboard reference to the film "Back to the Future 2"		
		
Scene 1: voice control activated garden from the ceiling. And man being strapped to ortho-lev machine upside-down	Scene 2: Boy tastes the fruits	Scene 3: Family use the garden in the kitchen
		
Scene 1: food is removed from its package and placed in a machine and activated/timed via voice control	Scene 2: Food comes out larger, warm and cooked	Scene 3: Family eats together

Table 5: Storyboard generated based on the verbal narrative of participant T

4.3.2.5 Focus Group

The results from taking part in a focus group; we were a total of ten (10) participants of 4 females, 6 males , who come from all sectors, industries and professions; The phases of activities in the focus group include: Step 1: Warm-up; Step 2: Presentation of my project to all; Step 3: Group was divided into four groups for discussions (I facilitated in a group of three); Step 4: Assembled in the general group for reflection and discussion. My short 6 minute presentation was a good introduction to the context of my project for an alternative

kitchen design. This set the ball rolling for discussions in groups on eating lifestyles. Below are some shared thoughts on the existing use of kitchen space (Table 6).

Kitchen activities	Notes
Cooking activities	"Averagely cook 5 to 7 times a week, cooking with spouse"; "creating something", "Cooking is daily meditation," "I am not so good at recipes".
Unusual kitchen activities	"During the pandemic the kitchen was my working space", "I miss having a kitchen island to socialize with guests"
Outdoor eating activities	"In the summer house, there is this big barbeque, and everyone gathers there, big table, barbeque together".
Shared kitchen	"I am not used to shared kitchen"
Children in the kitchen	"They like to eat when they participate in the cooking"
Technology in the kitchen	"Having a drink together with friends, even though they are far away…without the ipad", "I am fond of recipes, if it can be streamed …without having to unlock the telephone"

Table 6: Focus group notes

Discussions in the general group also highlighted the eating area as a centrepiece in relation to log fire based on the idea of sharing the cooking. Technicality of a flexible kitchen design solution was also explored. On social relationship, one of the participants highlighted that:

> "Relations to the farmers and food; where does food come from and how to integrate that to a kitchen? Relationships and how they are manifested to each other and connected to health. The kitchen from a cultural and historic point of view looks very dysfunctional".

The need for a dynamic/ flexible kitchen with a design to improve social activities in eating areas was recommended.

4.3.2.6 Questionnaire

The result of the questionnaire in (Table 7) includes 11 responses to open ended questions on alternative shared kitchen. 72.7% were male and 27.3% were female. 45.5% live with family, 45.5% live in single apartments and 9.1% live in a shared apartment. Their professions include architects, UX designer, computer engineering, Energy engineer, Industrial engineering, HR consultant, IT, Student Agriculturist and Business Administration. Below are responses regarding how their professions could improve social interactions, healthy eating habits and reduce wastage in the kitchen. Details of the questionnaire is in Appendix 1.

Profession	Social Interaction	Healthy Eating Habits	Reduce Wastage
UX designer	Food Events	Taking pictures of healthy dishes	Designing sustainable kitchens
Construction PM	Spatial requirements	Designing kitchen spaces that encourages healthy eating habits	Innovative domesticated technologies that recycle waste within buildings

Architect	Social responsive space	Educating clients in the design	Setting the correct amount of each ingredient for x person/s
Computer Engineer	Online booking website where 2 groups can use the kitchen for a specific period.	Monitoring the levels of sugar, vitamins and suggest what meals are recommended to cook	Improving storage methodology and minimizing energy consumption. Making cooking less stressful and time consuming
Energy Engineer	Hold interactive sessions and classes	Sharing healthy tips during every social gathering	By building recycling machinery
Working		As a maintenance technician, all equipment must be properly maintained	
HR Consultant		Paying more attention to employee wellbeing and general wellness	Reducing electricity consumption
IT		Schedule lunch time according to when I am hungry	Teach the essence of not wasting food
Agricultural Student			
Ind engr & Bus. Admin		By reducing the use of sugar and encouraging marketing	
Industrial Engineer	Monitoring the room	Offering fruits	Monitoring

Table 7: Results from Questionnaire

4.3.2.7 Iteration on Precedents- Architecture and Food

Earlier, we looked at precedents for the future kitchen. Here in this iteration section, I will be looking at the architecture in the Nordics and their design proposals related to food, architecture and the kitchen. I will also explore historical tropical architecture in West Africa and the relationship to the kitchen and climate.

Design with nature (McHarg I., 1969) takes into account a planetary, holistic vision, where city and urbanization are understood as a continuation of natural landscapes. Victor Papanek, Rachel Carson, Buckminster Fuller, and Jane Jacobs advocated holistic design and a focus on identifying the root cause of a problem, as opposed to redesigning or repairing a damaged system. Jacobs highlighted the need for a 'living neighbourhoods and street ballet' as a measure of a liveable and healthy city. The regenerative approach aims to create a more resilient and inclusive future (Krezlik, 2021). In making cities liveable, Gehl promotes pedestrianization, which focuses on public space and human interaction (Gehl, 2010). Below are some current design works by architectural companies in the Nordics addressing sustainable food production and architecture.



Figure 10: Oceanix city (Walsh, 2019)

Oceanix city by Bjanke Ingels Group is a response to the prediction that by 2050, 90% of worlds cities will be dealing with rising sea levels (Figure 12). The scheme enacts flows of food, energy, water and waste. They imagined a floating city designed to incorporate all the sustainable development goals into an ecosystem. Using energy for power (solar panels, wind turbine, wave energy converter, algae bioreactor, energy storage, heat exchange). Grow all the food locally- fish and plant based, deal with waste locally with community garden compost. Design to be modular, buoyant, designed to resist storm.



Figure 11: EFFEKT- ReGen Villages (Frearson, 2016)

The project – ReGen imagines a community of buildings that produce all their own food and energy – a model that aims to tackle a wide spectrum of global issues, from the food and water crises to the rise of CO2 emissions (Figure 13). With ReGen Villages, residents become part of a shared local eco-system, so different families can take on different roles in the community. Each village would include a series of public squares, integrated with technologies like electric car charging points. Individual homes would integrate photovoltaic solar panels to generate power and heat water. They would also feature passive heating and cooling systems, as well as natural ventilation, helping to keep electrical demand low. There would also be closed-loop organic food and waste systems.



Figure 12: Thinking Outside the B(l)ox, Hele Landet (Neighbourhoods for Generations, 2023)

Thinking Outside the B(l)ox: This attempts to address the problem of how we can bridge generations in urban areas. They created a pilot project called “Urban Farming Restaurant” that: Embraces different means of contributing; Engages people with different perspectives and at different levels; Empowers people socially, organizationally, environmentally, physically and economically. The project provides a range of opportunities for involvement, catering to diverse groups with various engagement options that suit individual means and preferences (Figure 14). It includes cultivation of agricultural land in urban spaces, running commercial business in housing association’s area (rooftop, outdoor spaces, etc) engageing tenants through jobs, free communal dining, access to healthy organic vegetables. An example of this framework is Botildenborg social farm and kitchen. Through food cultivation, they contribute to job training, internship programs in social farming, language training, school gardening, community gardening, intergenerational and intercultural (seniors and children) activities around food.

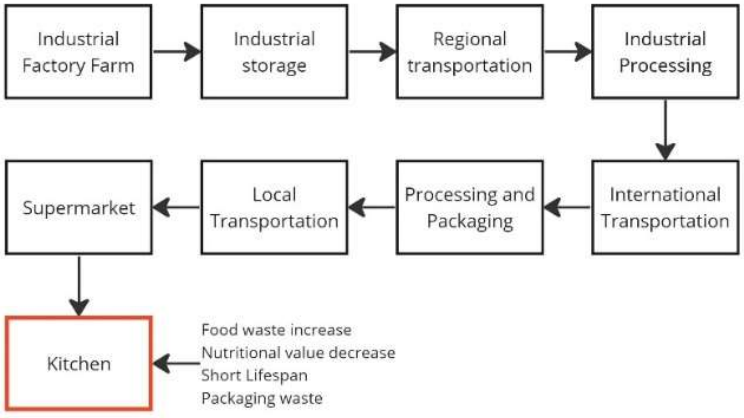


Figure 13: Current process of food production (Neighbourhoods for Generations, 2023)

There is an immense amount of waste and hazard in our current process of producing food (Figure 15). The hazards include; increase in food wastage, reduction in nutritional value, harmful pesticides, low paid labour, biodiversity loss, energy consumption, greenhouse gas emission, water wastage, plastic consumption, stimulated overconsumption, unequal pricing. Thus, the kitchen has a short lifespan of food, packaging waste, nutrition value decreases and food wastage increases.



Figure 14: Feedback to Høje Gladsaxe (Neighbourhoods for Generations, 2023)

Høje Gladsaxe neighbourhood: This is an idea of how a circular food and waste system could bring a neighbourhood's generations closer (Figure 16). Examples of initiatives relating to innovative food systems can be seen in the Høje Gladsaxe neighbourhood in Denmark. This includes; Lars Fairytale Garden where residents with garden space grow flowers and vegetables; Høje gladsaxe food bank – food from district are collected, frozen and prepared for community events; Green Neighbour expands awareness about sustainability and climate change; Food waste group- volunteers focusing on minimising food waste; Høje gladsaxe eating together- volunteers prepare monthly food for families church event; The food and Film Club about food preparation and communal dining activities; The Nature Detectives - learning from natural surroundings.

Tropical architecture: Maxwell Fry and Jane Drew were British tropical modernists (Jackson & Holland, 2016) with works in Nigeria, Ghana, India and Sri Lanka. While working in Britain, Jane Drew had extensive experience in kitchen planning in relation to health, hygiene and welfare. In Ghana, they proposed alternatives to some traditional compound houses. The traditional compound houses were made from adobe/cob, with multiple impluvium courtyard for collecting water, sunlight, social space, open kitchen, rooms for extended families and spaces for livestock and fencing (Figure 10). During iterations of proposals on the alternative compound houses, Maxwell Fry and Jane Drew incorporated indigenous needs for compound/ extended family living, open kitchen, rooms round the courtyard, flexibility and security (Figure 11). They also improved on the health, services and climatic factors.

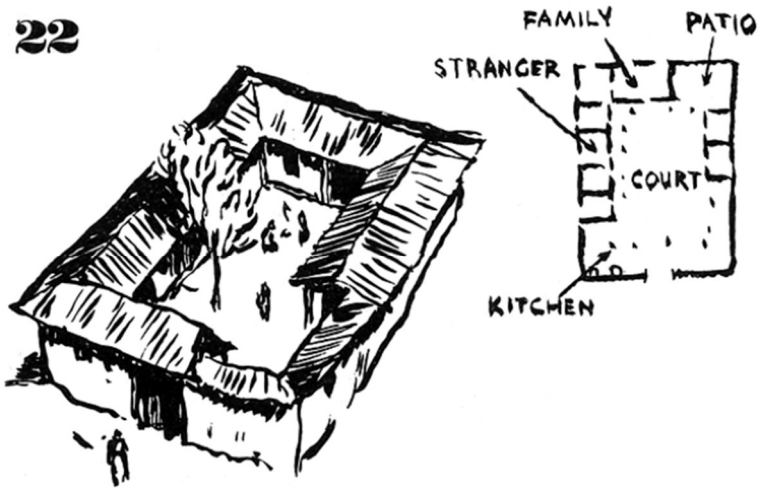


Figure 15: Fry's sketch condemning the compound housing type, 1947 (Jackson & Holland, 2016)

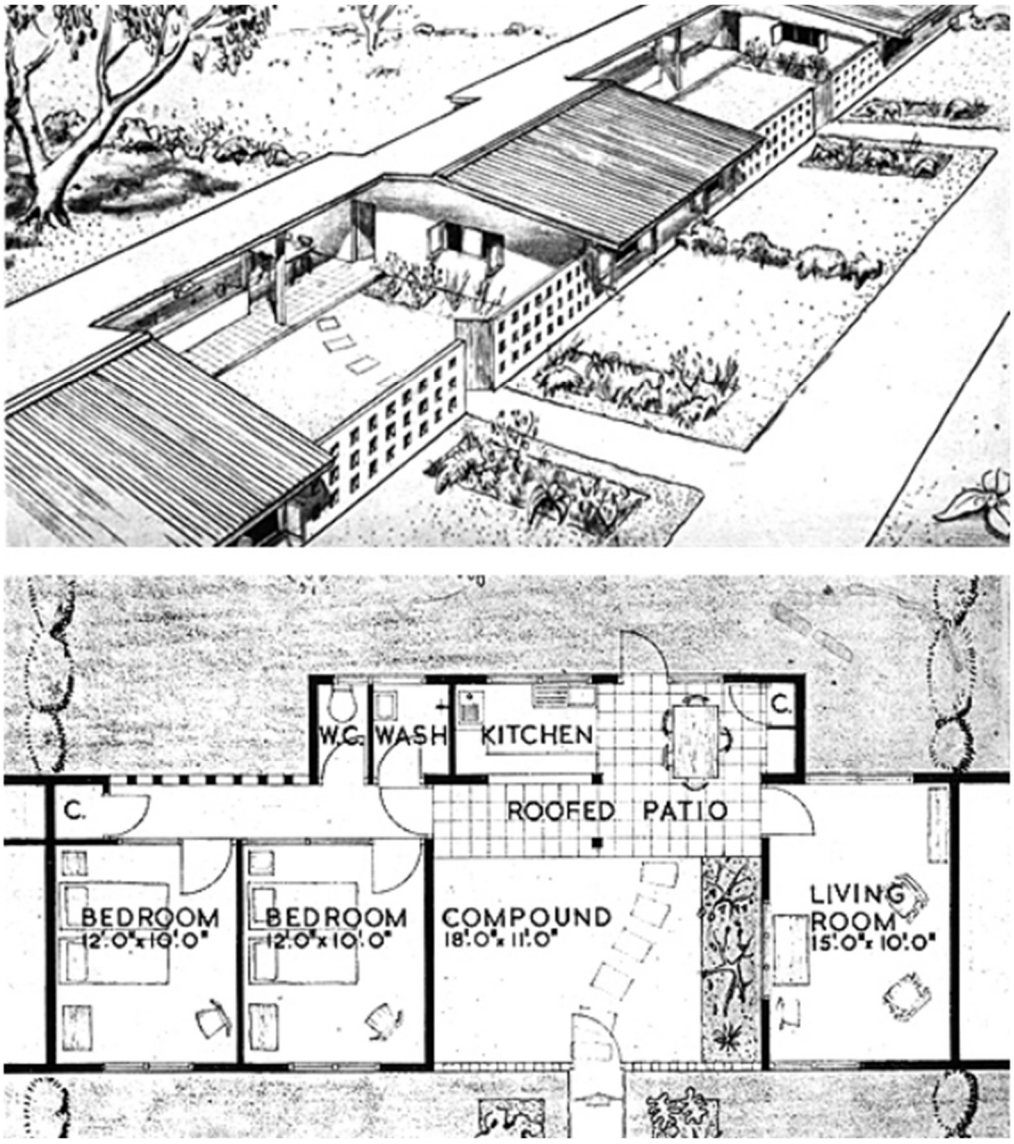


Figure 16: Initial compound design for Tema by Fry and Drew, 1956, (Jackson & Holland, 2016)

4.3.2.8 Prototype of a food event at Mälardalen University

In a course assignment in Spring of 2023, my colleagues and I developed a vision of a food festival for the University, and I executed a small prototype event. I was in charge of the planning and implementation of the event which created value by strengthening the place value potential for the students. We had cooking participants from Bosnia, Jordan, Italy, Palestine, Iran, Sri Lanka, and diverse visitors. It was an opportunity to meet new lifelong friends and future business partners outside of the classroom and learn new cultures and experiences at the event. The cooking participants co-created with friends in cooking and exhibiting their food and culture. The result was a diversity of students, teachers, visitors interested in intellectual, sensory and social experience at the food event (Table 8).

Intellectual Experience	Sensory Experience	Social Experience
		

Table 8: International Food Event and Mingle, Spring 2023

4.3.2.9 Observation of Housing apartment and a Choreography event

Observation of Housing apartments and a Choreography event (Figure 17). *Housing apartments:* Designed as private single rooms there are no common spaces. From the carpark leads to the security door which leads to the lift and the private ensuite rooms. Besides my course mates, it is possible to not make new friends in the housing unit. The second housing block has a small garden with an outdoor bench; however, it is rarely used. Social connections are dependent on already established social networks from home countries, school, work or WhatsApp groups, etc. Last Christmas, the landlord organised a xmas party/buffet, however there was a low turnout considering the number of building occupants (source: a course mate). This questions the motivation of tenants to a xmas party/ buffet to socialize with strangers-neighbours. It is possible that a co-creation process and spaces designed for organic interactions could bring about a different outcome; *Choreography:* the design of the dance was an intentional divergence and convergence of activities. There were the DJ's pulse beats, the choreographers dancing at the centre, the facilitator on the piano and mic explaining the process, the spectator's nodding to the beat, diverse people- old and young, able and physically challenged. There was a flow from bright stage lights to dim, flow from slow tempo beat to fast, from silent to loud music, and the stage smoke. There were speculative scenario dances, e.g how would you dance if you are one with space and see planets everywhere, how would you dance if inspired by sea waves and techno music, or the floor is made of lava, and you have special shoes etc. Towards the end, the audience were invited into a chain of movement and then loud blasting music and dance. I was drawn into a group, and I had so much fun.



Figure 17: Choreography and Student housing

4.3.3 Findings of Ethnography Study

Below is the summary of the data taken from the ethnography studies, participants experience of the future kitchen and food related activities. Grouping was made based on user needs, concept solutions and awareness of political structures (Table 9).

Groupings Kitchen Needs	Notes
Exploring User Needs Photo Study Storyboard Focus group Observation of housing Interview	Healthy food, wellbeing, exercise, socializing with family and friends, exercising, security, sustainable living, being close to food sources were some users' needs identified. A dynamic kitchen space that can be used for multiple uses was also a need. Observations of the private housing apartments shows how design of buildings could affect people's behaviour. Moreover, it shows that some tenants need more motivation aside from free food to attend social events organised for them.
Exploring Concept Solutions Precedents Questionnaire Interview Observation of Choreography	The current food system causes bad health, social inequality, dominant bonding networks and environmental hazards. The precedents showed many design proposals for the future kitchen and the ecosystem of shared kitchen for food production and upcycling waste. The questionnaire looked at how different professions could address issues of social integration, food wastage, functionality and eating habits. This section shows the need for bonding and bridging in neighbourhoods by directly involving people through mutual understanding and different motivations.
Awareness of Political structures Prototype of a food festival Precedents – Tropical architecture	Self-organization and co-creation in food festival was experienced where different group of friends prepared meals for the food event. The participants were motivated by the experience of the diversity of people, cultures, taste, and sound. They used their time and resources to make the event a success. As the project manager with low social network, I had to be diplomatic as people in common social networks mirror each other (i.e in inclusion/exclusion, invites, attendance, support and contributions) and dominant networks pose a risk of dividing and fragmenting groups.

Table 9: Grouping Ethnography studies and Kitchen needs

4.3.4 Discussions on the used methods

The following will discuss the learnings of the methods used in this thesis project.

Precedents: the precedents helped as a source of inspiration in the search for future kitchens. Elements of sustainability were captured in the home environment. It also highlighted the gaps and area under research in terms of alternative shared kitchen environments. *Semi-structured interviews*: the interview was a means to get knowledge from users/people about their life experiences using the kitchen. The interviewees had different experiences to share. Two participants highlighted good experiences while the other highlighted some design issues because of his design background. *Photo-studies:* the photo study made the participants' narrative visual, and I could empathize with his needs. *Storyboarding:* the storyboard in the form of film snapshot was interesting in eliciting past reflection from the participant. The participant could recall a movie that he watched years ago and present it as his idea for a future kitchen. In my reflection, it would have been good to get new ideas that could be added to the thought process. *Focused group studies:* The focused group was a great method to hear from professionals and natives of the country. Information brought new insights and perspectives for my thesis project. The use of welo platform (which is different from the usual zoom meeting; such as- new virtual environments like a stage for presenting with audience seating, the coffee rooms, etc) was interesting to participate in a new virtual environment. *Questionnaire:* open ended questions are not participants favourite as it requires deep thinking compared to close ended questions with solution selection. I was able to have a diversity of eleven professionals participate. *Prototyping:* My experience from organising a food event was relevant in this thesis as some food system initiatives proffer an event as a good way for socializing in neighbourhoods. Observations: My lived experience as an international student enabled immersion into the context and culture of the socially disconnected. Ethnography- generally the ethnography studies for collecting data are core to the innovation process. I engaged with the participants to receive information and understand their needs at multiple levels and meanings of life-worlds. The learning style in the ethnography study is the diverging style. I observed concrete situations from different points of view.

4.4 Frameworks

At this stage of the process, a lot of data were processed while simultaneously identifying the gaps for users and consumers. It also involved recognising and challenging any presumptions that I may have had on the results, as well as differences in the beliefs and preconceptions. The main goal of the framework stage is to reframe the problem, either by creating a new narrative about how the participants can solve it or by developing an alternative perspective on it. This will enable the team—the participants and me—to generate new ideas for solutions. Framing is essentially the process by which I look for compelling narratives within the vast amount of data gathered, themes/ patterns of behaviour among the numerous behavioural occurrences that were noted, and gaps in the system of use, usability, and meaning that defines the innovation or solution (Beckman & Barry, 2007).

Application of developing frameworks in this thesis: I identified interesting stories from data gotten in the ethnography methods. I questioned assumptions that I initially had, of reinstalling the kitchen with emerging technologies. In order to develop a narrative about how to design an alternative social kitchen that will be useful for people, I identified interesting stories made by the participants, and interesting dimensions of users behaviour and created a simple model.

4.4.1 Findings of Framework

In order to extract information from the phenomenological analysis of the ethnography data, I identified interesting stories such as the need for healthy food, relationship with food source, social space in the eating area, older adults not keen on shared kitchens in buildings etc. Key codes were the user needs, political structures and kitchen concepts. Below are the dimensions / codes identified (Figure 18). This represents thinking patterns in a nonlinear visual way.

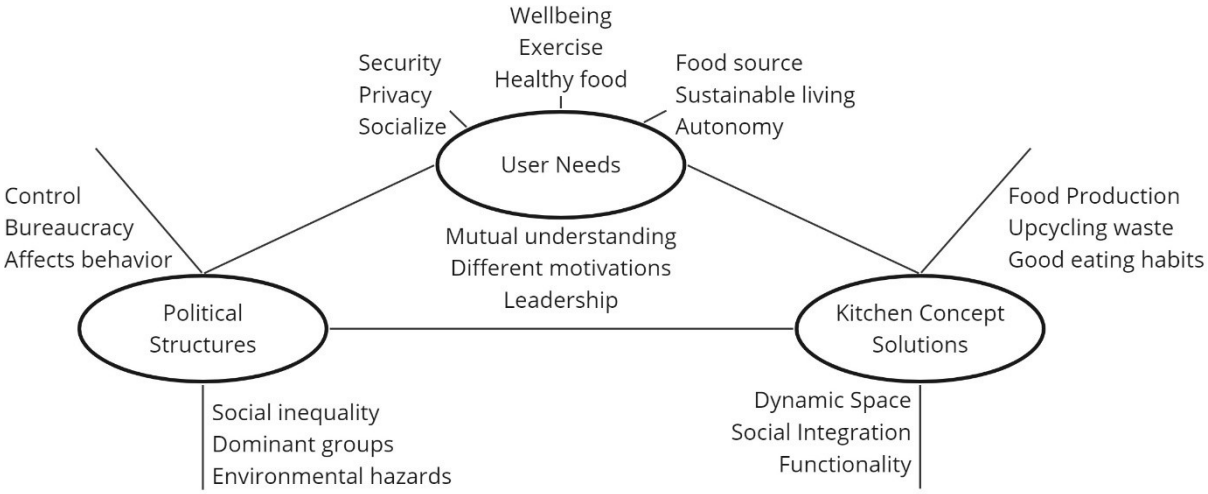


Figure 18: Design Codes

The precedent projects show different attempts at designing a future kitchen; however it has not been adopted at a global scale and this is due to the need for a core understanding of people's needs in a systemic way. Themes were discovered (Figure 19) based on phenomenological analysis and other actors connected to the kitchen. The interview, storyboard, focus group, photo studies and observations gave us insights into the user's needs i.e,- what people like or don't like about the kitchen, their lifestyles/habits and need to stay healthy. The precedents, focus group and questionnaire gave more depth by;- seeing the kitchen as relational to people, farmers, culture; the complexity of needs and relationships; and how people would like to gather around the heat and have a dynamic space in sharing and cooking.

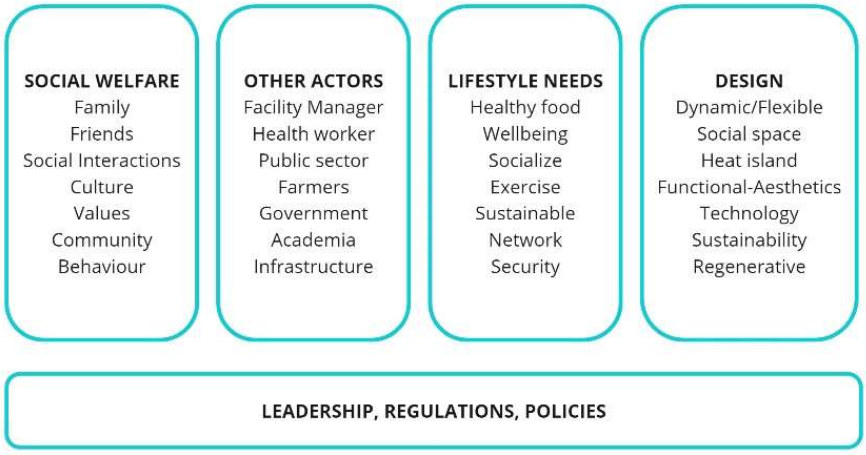


Figure 19: Design Themes

The three key findings in the phenomenological analysis includes: exploring user needs, exploring concept solutions and awareness of political structures. The design themes discovered were social welfare, lifestyle needs, design and other actors involved in the food system of the kitchen. This is summarized as follows:

1. Exploring user needs: this involves lifestyle needs and social welfare.
2. Exploring concept solutions: this involves design solutions required.
3. Awareness of political structures: this involves other relevant actors in the ecosystem.

To address the three findings for kitchen needs, the perspectives on innovation processes are employed (Garud et al., 2013; Tidd & Bassant, 2013). The emerged ideas are to be implemented within homes, across multi-party networks, and within neighbourhoods. The kitchen innovation implies the implementation of an improved service, a new marketing method for housing development. This will involve changes in kitchen product, process, context and the underlying mental models in the urban setting. Below in (Table 10) are the dimensions for innovating neighbourhood design using Sawhney et al. (2006). The innovation process will require leadership that can harness rather than minimize innovation in complex and ever dynamic organisations. Organisational and contextual ambidexterity (Garud et al., 2013) uses both exploration and exploitation processes in appropriate cultures and structures.

Dimensions	Definitions	Kitchen Need
Offerings	Develop an innovative new service	Exploring user needs: this involves lifestyle needs and social welfare
Customers	Discover unmet tenant/building occupant's needs	Exploring user needs: this involves lifestyle needs and social welfare
Processes	Redesign core operating processes	Exploring concept solutions: this involves design solutions required
Presence	Create new distribution networks	Awareness of political structures: this involves other relevant actors in the ecosystem.

Table 10: Dimensions for innovation

A conceptual model (Figure 20) was developed based on the emerged ideas to be implemented within homes, across multi-party networks, and within neighbourhoods. The model shows the family at the centre connected to food industry, urban and health sectors for wellbeing through relationships with the kitchen, social activities and farmers, with right policies by the government, and public sector. The kitchen design is connected to systemic, regenerative and speculative sustainable design.

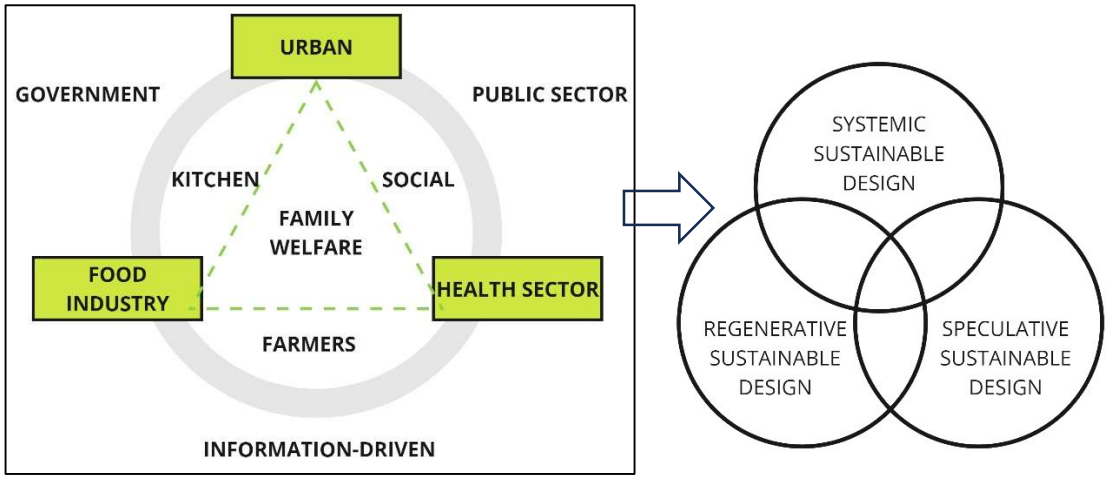


Figure 20: Concept model for the kitchen environment

Describing the conceptual model in connection to service logic and value co-creation (Fujita et al., 2018; Garud & Turunen, 2021; Sawhney et al., 2006).

- Family welfare: at the center, value is experienced by the family members (user value).
- within homes: the inner circle shows the family wellbeing (user experiences) within the kitchen environment, having social interactions and connected to the farmers.
- across multi-party networks: The liveable urban, food industry and health sectors (user resources) are interlinked and enable the inner circle.
- and within neighbourhoods: the government, public sector and information driven care (organizational resources) are enabling the innovation process in the ecosystem.
- The innovation dimensions of offerings, customers, processes and presence; guides companies to harness complexity and add value to the kitchen using systemic, regenerative and speculative sustainable designs.

4.4.2 Design Concept

The phenomenological aspects of the kitchen can be described based on the three-dimensional method (Fairclough, 1992), this include: text (the objects of linguistic analysis), discursive practice (the production, distribution and consumption of texts), and social practice (Ideology and power are created at this level, which has consequences for how life is lived, and society is structured). In this section, based on the design codes of the ethnography methods such as: user needs, kitchen concepts and political structures, the kitchen will be discussed under "the open home as a social catalyst", "the kitchen as the heart of the open plan" and "the open home as stage".

The open plan housing as a catalyst for social life: This is connected to user needs for security, privacy, socializing, wellbeing, exercise, healthy food, connection to food source, autonomy and sustainable living. As mentioned in the focus group, the benefit of the open plan housing with a kitchen island, is that guest/ visitors do not feel isolated when the host is cooking. The guest can join in the cooking or the conversation. Festive activities in the open living space can extend to an outdoor deck, garden or swimming pool area. Also, everyday socializing with family is made easier and children can learn to be helpful and responsible in the kitchen. Different family activities (watching TV, cooking, doing homework) can occur concurrently in the open plan housing (Figure 21). The image of the social housing inhabited by a happy family with a lot of socializing is, however, an ideal image that often does not reflect reality. As mentioned in the focus group and precedents, there are many single households where the tradition of socializing is dysfunctional.

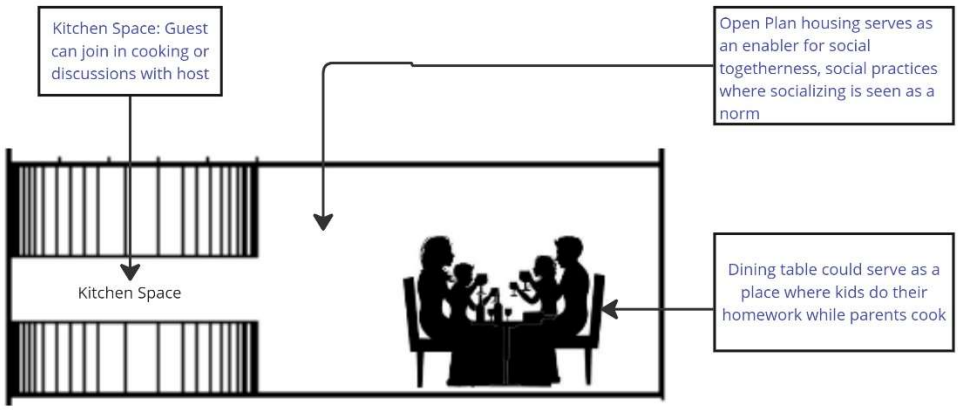


Figure 21: Open plan housing

The Kitchen as the heart of the home: This is connected to kitchen concepts such as food production, upcycling waste, good eating habits, dynamic space, social interaction and functionality. The precedent on future kitchen shows architects, interior designers and housing developers' interests in unique and sustainable kitchen solutions based on people's consumption habits, culture and sense of belonging. The functionality of the kitchen entails ease of work, adequate work surface and storage. The open / vanity kitchen also serves as a form of displaying social status, where the kitchen is aesthetically appealing with appliances, furniture, finishes and good taste is showcased. Examples from the questionnaire on

functionality shows that aesthetics, social features, equipment, process of cooking, workspace are important to kitchen users. Cooking is a way of living and looking at the world, beyond the basic need to satiate hunger. Food and cooking are related to the four roles that the kitchen must play: that of a social gathering place, a workplace, a display object, and a symbol of security (Figure 22). The importance of having a relationship with the food source was emphasised in the focus group. People's perceptions of the kitchen and the significance of food in life are influenced by their social and cultural identities as well as their economic circumstances. Without any cooking, an empty kitchen is ineffective as a symbol.

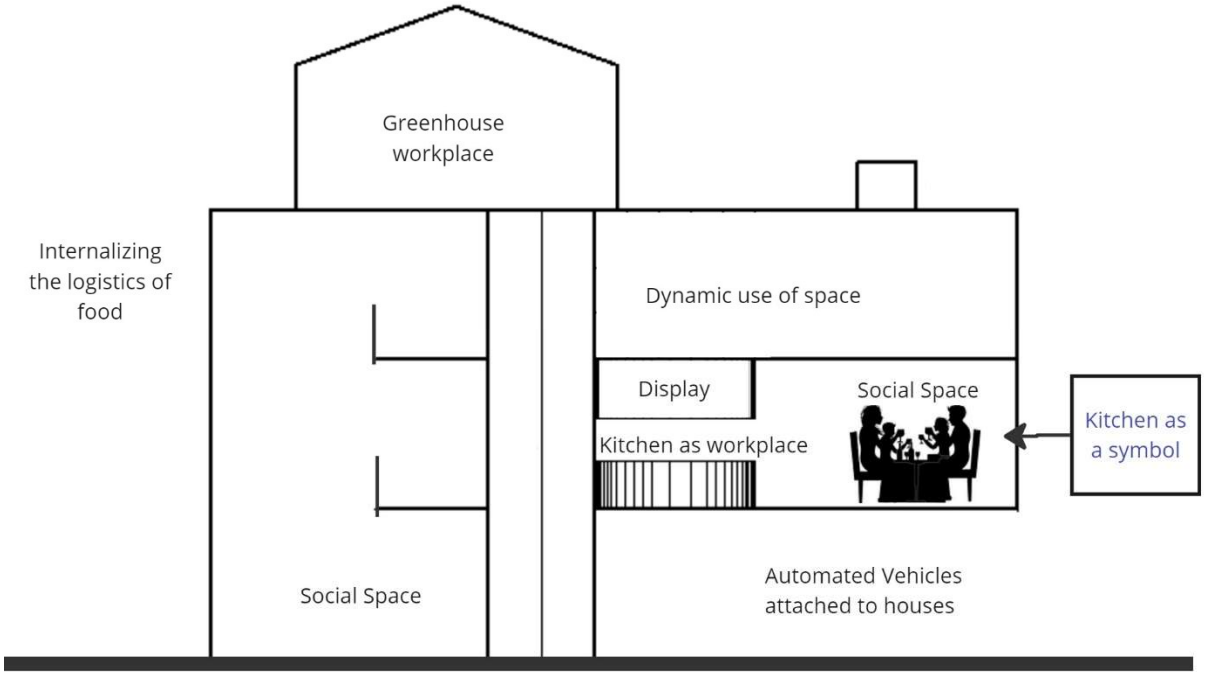


Figure 22: The kitchen as the heart of the home

The open plan housing as a stage: This includes political structures such as control, bureaucracy, effect on behaviour, social inequalities, dominant groups, environmental hazards. The open home functions as a stage for self-presentation of identity. Boundaries between the outside and inside are enabled with large windows, balconies, deck, furniture and garden. The openness demands that the space is kept tidy for guests or outsiders looking in. The visibility functions as both a means for showcasing and conforming to societal norms and idealist behaviour. The precedents show that architects are interested in creating a vision for good life, good taste in housing neighbourhoods through sustainable practices of reducing food wastage, renewable energy source, closeness to food source, technological innovations and social change (Figure 23). This is to address loneliness and unhealthy eating habits.

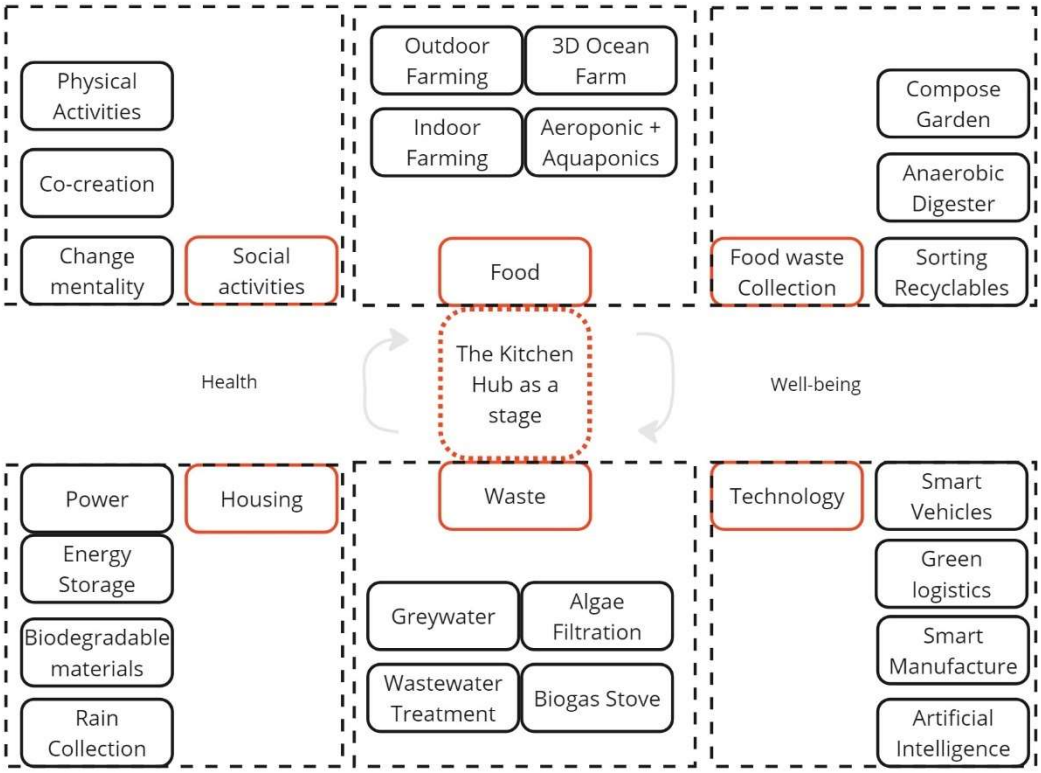


Figure 23: The Kitchen as a stage

General Discussion on Frameworks

Frameworks- involves the innovation process moving from the concrete to the abstract realm in order to make sense of all the data that I collected. Framing and reframing of the data to identify themes and identify a focus that has meaning for the customers/people. The learning style here is the assimilating style. Here I was more focused on ideas, abstract concepts of the information and creating a conceptual model/design. Value cocreation (Fujita et al., 2018) will be required involving actors and resource integration, enabled by service exchange, internally supported by institutional arrangements, thereby developing a nested service ecosystem.

4.5 Imperatives

Imperatives is a collection of specific requirements or guidelines—also referred to as design principles—that must be followed when developing the invention. From the insights and models developed throughout the framing stage of the innovation process, imperatives are taken out and made very evidently connected to a comprehension of user or client demands. A value proposition is an explanation of the observable advantages that users of a product or service would experience (Beckman & Barry, 2007). *Application of using Imperatives*: following the exercise of framing and reframing the user needs data, I then developed a value

proposition that must be met by the new concept. I designed a workshop using storyboards, and conversation cards to support participants in the workshop.

4.5.1 Findings of Imperatives

The imperatives are design principles, or value proposition—that provides a very high-level specification for the design of a shared social kitchen. They provide a guiding vision for the remaining activities in the innovation process. Creation of the imperatives often represents the first major point of convergence for the innovation team; prior to the development of imperatives, I have been gathering and analysing data in a highly divergent and exploratory mode. Earlier in the reframing phase, I explored understanding the situation of the social kitchen and highlighted the core knowledge regarding the design challenge. I discovered multiple interpretations and suggestions for an alternative kitchen, and concluded that creativity is needed to move towards a solution. A creative climate in an enabling environment is required for emergent innovations. Below in (Figure 24), I prepared a workshop plan by making it concrete and actionable.

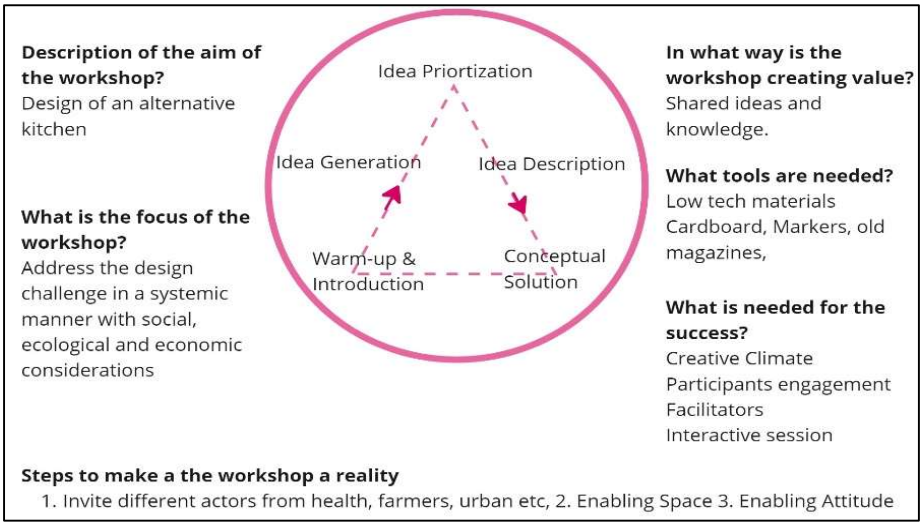


Figure 24: Concrete workshop planning for this thesis

A workshop was conducted using a speculative approach to gather a wide range of concept generation techniques and come up with an alternative solution to the shared kitchen design.

Speculative design/Design Fiction**:** Speculative design approaches use design to imagine future states of affairs and speculate about how things might be. Design fictions contribute to the speculative design methodology by assisting in the creation of artefacts from various possible near-future worlds and demonstrating various design approaches based on user needs (Bleecker, 2009). In order to understand the needs and expectations of users, artefacts are used to communicate with them (Çetin Er & Özcan, 2022).

4.5.2 Speculative Design Fiction approach

Hebrok & Mainsah (2022) highlighted some definitions of design fictions by other researchers that I found useful for this thesis. According to (Sterling 2012) design fiction is characterised by the intentional use of diegetic prototypes that allow for the idea of change. Also, Dunne & Raby (2013) defines science fiction as a medium "not to show how things will be, but to open up a space for discussion", with the combination of design, science fact, and science fiction (Bleeker, 2009). Design fictions, as noted by Hebrok & Mainsah (2022), creates fictional alternate worlds that make sense of the artefacts that make them up. They do this by utilising any and all media. These artefacts frequently include comic books, films, fictitious documentaries, interactive academic texts, and tangible prototypes like catalogues, newspapers, and user manuals.

According to (Fu & Zhu, 2020), conceptualising, assessing, and reflecting on potential future scenarios and product ideas is made possible by design fiction in various physical settings, technological contexts, and different users. Furthermore, this unique approach has drawn the attention of social issues for future research due to its comparison with other qualitative research methods (Fu & Zhu, 2020). This is because it focuses specifically on the social aspects of human context and the viability of emerging technologies. rather than "technology-driven" and "enterprise-oriented" ideas for other product design approaches, with "situational" as the main design outcome. Because of this, instead of concentrating on the non-textual conceptual products and services, stakeholders can become fully engrossed in the story of the design novel. Accordingly, a first-person subjective viewpoint can be used to analyse the social aspects of design proposals (Fu & Zhu, 2020). Physical personas were used during the workshop to serve as a starting point for the design fiction, allowing the imagination of future possibilities for the relationships between people, needs, lifestyles, food, other actors, kitchen spaces, social interactions, technologies, health, and wellbeing.

4.5.3 Method

I conducted a design workshop to observe how participants use artefacts (cards, kitchen model, kitchen appliances, persona) to generate ideas and develop sketches from them. The participants were designers and non-designers. The rationale for the selection was to observe group interactions and idea generation. I acted as the researcher and the facilitator. The conversational cards were used to support the participants in idea and concept solutions, while imagining contexts and scenarios in brainstorming and sketching (E. B.-N. Sanders & Stappers, 2008) of an alternative social kitchen design. The artefacts were used in the workshop and timing was scheduled for forty minutes. The interactions were recorded, observed, and notes taken. A critical approach was used to analyse the result.

4.5.4 Designing the card template

Design of cards: Using a future vision generation process (Fu & Zhu, 2020) includes the following: 1) The world outlook, 2) The theoretical foundation, 3) The new world, 4. Roles, 5) Scene, 6) Props and metaphor, 7) The system diagram. This results in the design of conversational cards for storytelling and imagination based on the real context (Table 11).

Elements	Vision generating process	Design
The world outlook The theoretical foundation	Generate Future Chronicles	Sustainability in the future-environmental, social and economic aspect is very important for health and wellbeing.
The new world Scenerio Persona	Generate Future vision from the chronicles	Draft example: We are living in the same apartment, and we will share the same kitchen, in that kitchen we will develop …….
Props and metaphors System map	Vision shaping	How will you design the kitchen space that is self-sustaining your lifestyle and allowing social interactions?

Table 11: Process of card template design

The cards were used to help the participants imagine playful and thought-provoking narratives of future sustainable food systems (Figure 25). The cards were designed to trigger the imagination of participants in co-creation workshops, whilst at the same time narrowing the scope down to certain contexts. Low tech materials tools were used such as cardboards, markers, sticky notes and old magazines. The approach was designed to prompt reflection on changing lifestyles, future needs and reflections on the environmental, social and economic aspects of sustainability.



Figure 25: Designing the template cards

4.5.5 Designing the process of the workshop.

Workshop stages include; Step A: warm up and introduction; Step 1: concept generation/ Naming; Step 2: Framing - Sketching scenarios and concepts for future social kitchen Step 3: Solution- Reflection (Figure 26).

Challenge posed “How can the kitchen through food be used to improve social interactions?”.

This step is facilitated by a visual tool posing three questions about the future: Who or what affects the area? Is there any uncertainty in the field and in general? What are the current trends. These questions establish the basic framework of the scenario and help the mind become open to the particular theme. This is done individually.

- First, the team members share their ideas and perspectives about the future to start the brainstorming session. As ideas and information come to mind during the brainstorming, they are encouraged to build on each other's work. About 20 minutes into this step, the groups are asked to concentrate on one scenario or problem that most interests them so they can work on it further.
- The second phase, which is framing, all teams are required to use a storyboard to illustrate and narrate a story about the situation they have been focusing on, as well as to record their narrative. All the teams are then asked to concentrate on a single, distinct section of the story to develop their concept at the conclusion of this step.
- The last step creates conditions for others to understand the concept, and it also makes the concept concrete and helps bringing the work forward.

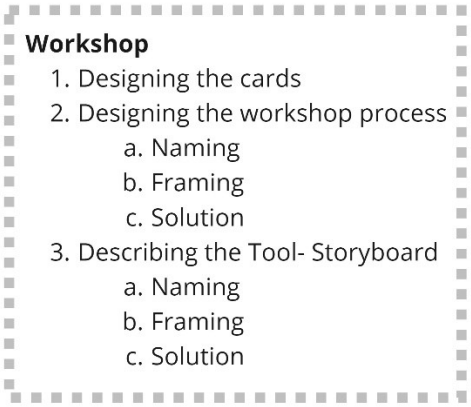


Figure 26: Workshop design

4.5.6 Describing the Tool used in the workshop - Storyboard

Storyboarding: We mentioned earlier that in the second phase of the workshop process, the participants will use a storyboard in narrating their story. This section is describing storyboarding as a tool. Schön (2016) defined the professional practice by incorporating processes of reflection and learning. In practice, reflection involves both thought and action.

The designer "shapes the situation in accordance with his initial appreciation of it; the situation responds to his talk-back; and the two engage in a reflective conversation with the situation." Reflect-in/on-action is a problem-solving methodology that the designer employs. To solve ill-structured problems, one must first structure the issue at hand before proceeding to solve the well-structured problem. This process entails reframing the problem, which consists of both a convergent and divergent phase. The theory, as formulated by (Schön, 2016) provides an analysis of the ways in which professionals themselves engage in driven by action thought. The model facilitates comprehension of the designing process iteratively, including naming, framing, moving-reflecting, and back-talk. Storyboarding can be conceptualised as a form of communication, akin to how a story is told in narrative theory (Wikström, 2013). The direct content of the linguistic in a narrative discourse is focalization. The focalizer is distinguished from the narrators by their connection to the verbal narration and the iteration during the process of storyboarding. Storyboarding is utilised to generate insights and gather data regarding workflow processes. Metaphors, lines, text, and symbols of objects serve as narrative focal points in the storyboards (Wikström, 2013). The design process will be examined, including naming, framing, moving-reflecting, and back-talk.

Discussion on Imperatives

Imperatives- The learning style for designing the workshop is the converging style which are abstract conceptualization and involves active experimentation. I applied the needs of the users in the planning of the workshop towards using qualitative research, speculative design and enabling space to propose future scenarios.

4.6 Solutions

This entails selecting and testing the best option that satisfies the requirements, design principles, and rules specified in the imperatives stage (Beckman & Barry, 2007). A diverse array of concept generation techniques will be employed to produce alternative solutions grounded in the imperatives, which are inextricably linked to ethnography-based methods. I will then select which solutions to advance using a set of concept selection techniques that are well-documented, and I will collect feedback from potential users through a variety of mechanisms.

Applications of the solution phase: idea creation, idea selection, and idea testing will be explored in the workshop. Techniques for generating concepts can be either intuitive or rational. Divergence and convergence of concepts are examples of logical procedures. The many brainstorming methods (e.g., group, individual, drawing, and word association) are examples of intuitive procedures. The participant will choose one idea to execute. Lastly, concept testing will provide feedback for choosing ideas or for building a prototype and testing it with people.

4.6.1 Findings of the Solutions phase

Result of the Workshop

Persona: Here is introducing the persona "Different". He/she looks, speaks, talks, dresses, acts differently from others. Different is not specific to any country/culture/tribe/age/sex/health disability or race (Table 12).

 "Different" = the Golden Balloon	• In the bus, Different has the window seat. People would rather seat alternative seat from him. Not directly in front or behind. • At the bus stop, if Different is standing or sitting at the bus shed, people would rather stand outside of the bus shed and avoid eye contact. • Different has been living in an 11-storey accommodation building for 5 years, and still has no friends. • He sits alone in public spaces. There is a 5-meter distance from the next person, facing away. • Whatever the seating arrangement, people have reasons to leave his table or row of seating. He counts 10mins, and they are gone. They could even boycott the event or gathering entirely. • When Different sends out an email, people assume he is asking or will eventually ask for a freebie/ charity/ free pass. So, their response or no response is disconnected. • Others are surprised to see Different as tenant/customer in the building lift, or as manager/ owner, as a designer, as a leader

Table 12: Persona – "Different"

A song by Radiohead, titled "Creep" perfectly describes the mood and persona of "Different".

"….I wish I was special, you are so very special. But I am a creep. I am a weirdo. What am I doing here? I don't belong here. Ooh she is running out, she is running out again….."

I conducted a workshop with a co-design perspective to observe how participants generate ideas in an Enabling Space. The workshop was conducted with four (4) students divided into two groups (Group A and B). Group B was involved in the idea generation phase only. The role of the participants was to relate their experiences with food, kitchen, other users of the kitchen and ideation of improving the kitchen with technology. I created a persona "Different" to bring to focus issues of social inclusion. I also created a kitchen artifact and conversation cards for group discussion and idea generation. Sticky notes, pen markers were provided for each participant. The participants were told to bring a kitchen appliance to the

workshop. A four-step workshop lasting 40 minutes was designed. This include: warm-up, idea generation, idea prioritization and concept solution (Figure 27 and Table 13).

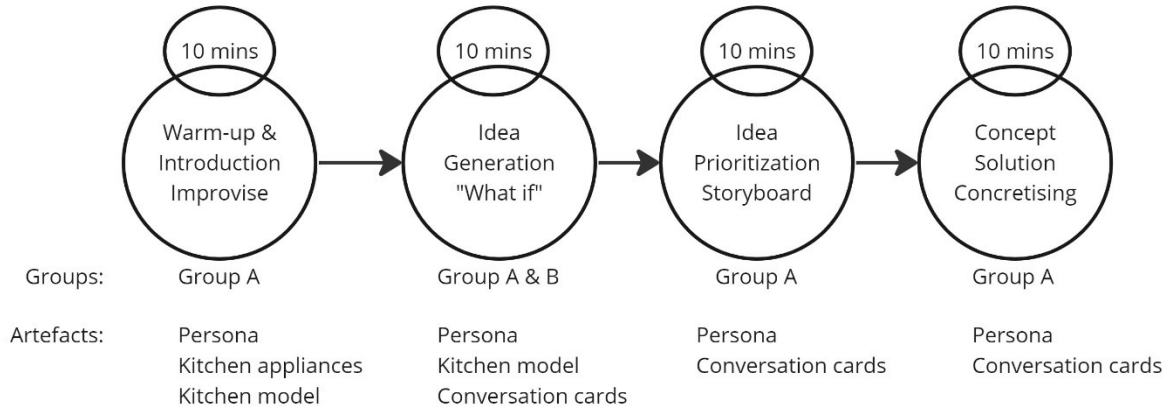


Figure 27: Workshop process

The workshop challenge: How can the kitchen through food be used to improve social interactions?

Steps	Activities	Result
Warm-up	What type of housing do you live in?	Two of the participants live in a shared apartment (ensuite bedrooms, shared kitchen); while the other two live in a private apartment. Issues of cleaning the kitchen sometimes arise in shared apartments.
	How have you felt socially included generally? How have you felt socially excluded?	Participants mentioned that they feel included in their apartment, at work and during group assignments in class. However, one participant mentioned he feels excluded in the shared kitchen apartment because of the type of smelly food he likes to cook.
	Why did you choose to bring this particular kitchen appliance?	They had cultural and sentimental reasons. Pictures were taken with the appliances after the warm-up stage.
	What are the future trends in your profession?	AI in determining price. Connection between mortality and climate change Smart manufacturing Digital twin
Idea generation	What if you are in a neighbourhood island with different people such as merfolk, how can you bring about social interactions?	Group A: Organising dinner, inviting people for lunch/dinner and asking if they have an allergy. Organise a workshop/information sharing on agricultural activities and discuss practices that can improve yield and health. Identify a common interest for food conversations, such as different ways of making potatoes, gardening, ingredients etc.

		Group B: people interact through smell. Iranian food requires patience, time and good appliances. The recipes are also difficult: Technology can help with online teaching, explaining the right alternative ingredients that can be used in different countries. Food sharing can help to reduce waste. Reuse of food in other ways can also reduce waste. E.g leftover rice can be baked to make Tahdig.
Idea Prioritization	Participants were told to choose one idea and generate a story	Group A chose to do a storyboard of their first idea: Organising and inviting different people for dinner
Concept solution	Participants generated solutions to make the idea work.	Concretising their concept to make it work.

Table 13: Workshop process

On future trends, the participants highlighted the following; Artificial Intelligence in determining prices, for example a software that will automatically give real-time prices to insurance products; Connection with mortality and climate change-such as how the weather is affecting the health of individuals and crops. Changing climate and health requires new mortality tables to price for insurance products. The participant highlighted how smart manufacturing is moving towards digitalization, smart maintenance, AI, simulation, extended reality and use of Digital twin.

It is interesting that Group B mentioned how the smell of food could bring people together while a participant in group A mentioned how preparing his cultural dish in his shared apartment led to people throwing open the windows.



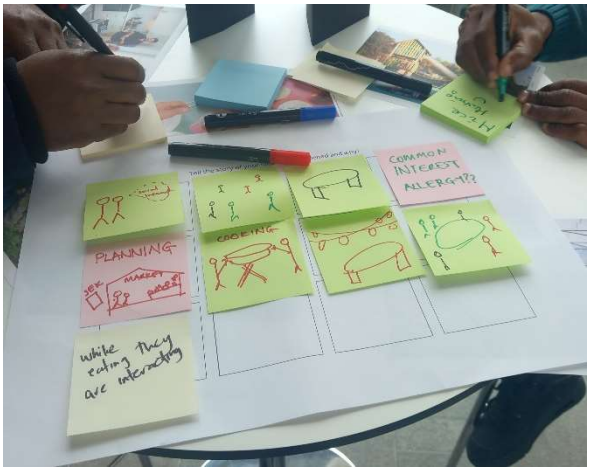


Figure 28: Group B Idea Generation; Figure 28b: Group A: Idea Priortization

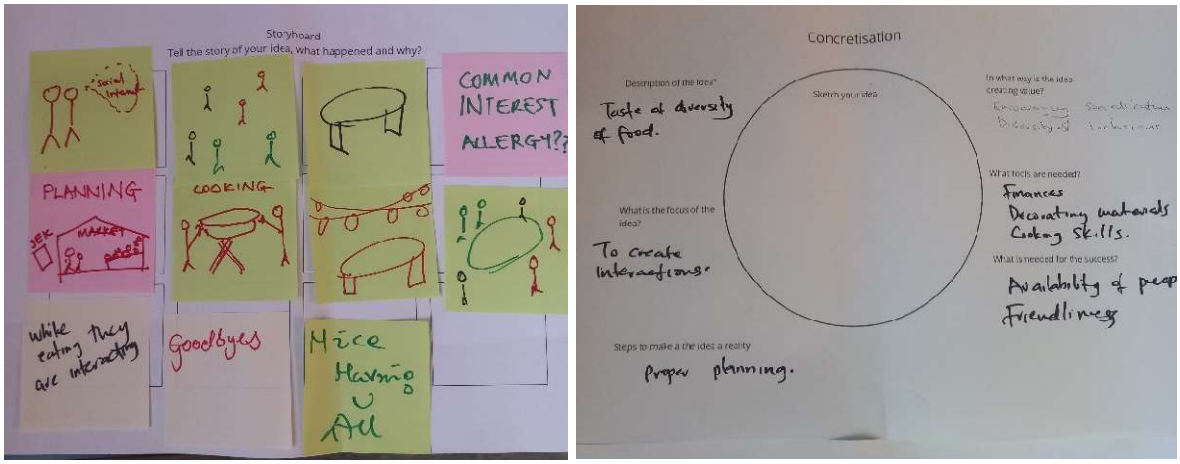


Figure 29: Storyboard and Concretization

Group creativity at the workshop: Group dynamics is a central process of group creativity that emerges from the interactions of group members in the production of new ideas and the ability to integrate ideas into a solution. Following the GroPro model, I as the facilitator was able to organise the participants, enable creativity and direct the group process (Backström, n.d.). The participants showcased their individual professional autonomy and self-efficacy during the warmup and idea generation stage (Figure 28). At the end of the warm-up stage, the participants improvised with their kitchen appliances by self-organising (Backström & Söderberg, 2016) in a playful body demonstration. They had an open attitude and did not think it was a gendered task to perform. Improvisation was their highlight of the workshop as they reminded me of the appliances they brought and were eager to showcase and tell their stories. The warmup and idea generation stage provided divergence of individual ideas and exploring new information such as mentioning future trends such as - the connection between mortality and climate change, smart manufacturing. Working as a group, they narrowed down on choosing one idea into a storyboard that converged *into a common understanding of planning a social dinner based on people's common interests and allergies* (Figure 28b). The group explored how a concrete solution could be actualized with resources (Figure 29).

On the social structures; there was a diverse professional team representing individual lifeworlds and group understandings of similar experiences such as use of spices, reusing of excess food in other ways, kitchen experience and future trends. Job diversity has positive significance for group creativity (Hülsheger et al., 2009) in the individual and collective activities carried out.

As the facilitator, working with a small group meant that there were not as many ideas generated compared to working with a larger group. The vision of the workshop was clear and different artifacts were used to support the group. Group flow (Sawyer, 2008), openness

and an improvisational attitude was seen during the storyboarding where the participants had to build a storyline from start to finish. It consisted of the 'Yes and' (…so what happened next….), deep listening and moving ideas to create a meaningful storyboard. There was cohesion in the two groups involving a cooperative atmosphere which increases group creativity (Amabile, 1996; Olsson, 2023).



Figure 30: Participants Improvising; Figure 30b: Artifacts

Artifacts: Workshop participants choreographed with their kitchen appliances to build up a mutual foundation for the diversity of the participants (Figure 30). This was exciting for the participants as they were able to display their emotion while simulating the cooking process. According to Schein (1984), artifacts such as the kitchen appliance brought by the participants can reveal underlying assumptions, values, and unspoken rules. For example, the pestle held by the participant can be used to make "eba" or "amala" (looks like mashed pudding) and a bigger pestle can be used in a traditional social process of preparing pounded yam.

The kitchen artifact was modelled as a circular dining table with chairs and in the middle is the cooker hob/heat for cooking. It was staged in the workshop to stimulate interaction to understand participants' experiences and perceptions during the warmup session of their feelings of inclusion/exclusion in the kitchen. Other artifacts like the conversation cards were used to stimulate idea thinking. The physical representation of Different (as a Balloon) was used to emphasize/focus the need for social inclusion ideas in the workshop, as I and the participants could be swamped with other aspects of the design. All the boundary negotiating artefacts (Lee, 2007; Schaeffer, 2015) such as the appliances, model, conversation cards and the Balloon (Figure 30b) were used in an iterative process to bring the diverse participants/ideas together. This culminated in the creation of a shared understanding of different perspectives (Star & Griesemer, 1989) in the workshop.

4.6.2 Analysis of the workshop

An enabling structure was created during the workshop using different artifacts and guides for the participants to engage and for new knowledge/ideas to emerge (Table 14).

Enabling dimensions	Definition	Learnings
Architectural & Physical space	The space in which innovation and knowledge is taking place.	The workshop was done in a conducive lobby space with chairs and tables (low and high) and an outdoor view.
Social, cultural and organizational space	Social groups essential to bring about innovation and new knowledge	Group A includes 2 international students: one from Cameroon (Product and Process Development - Production and Logistics) and the other from Ghana (Financial Engineering). Group B include 2 international students from Iran- one studying Computer engineering and the other Business. The teams that participated were qualified to gather different perspectives.
Cognitive space	Relationship between individual and collective activities and interaction with the environment.	The participants had individual thought processes, they interacted in groups, interacted with their kitchen appliance, with "Different", artifact, and with the cards.
Emotional space	Triggers emotions, supporting processes of knowledge creation	Each participant talked about how they felt included or excluded. What cooking meant to them.
Epistemological space	Identifying and understanding the innovation processes	The participants were involved in idea generation, idea prioritization, use of storyboard and concretisation
Technological and virtual space	This involves the technological environment	Low-tech tools were used such as papers, markers, sticky notes, balloons, etc

Table 14: Enabling workshop (Peschl & Fundneider, 2012, 2014)

Dance Choreography and Workshop: It will be interesting to take a closer look at the activities in the dance-choreography and the workshop using the wheels of rituals (Christrup, 2019) which is aimed at creating a collective state through feedback on emotional states and making it a ritual that can allow transformations. The three-guiding principle of wheels of rituals include: attention to the world (drop judgements), attention to oneself, and empathy. In the choreography, individuals and groups can get in contact with embodied emotional states, be aware of emotional tensions, and create states of consciousness that promote synergy and mutual relationships of the participants (Christrup, 2019). Below are the stages that I noticed and experienced in the choreography (Figure 31).

Co-initiating- building common intent, willing to listen and empathetically understand the performance (Scharmer, 2007). Spectators decided to attend the Festern, I was curious about street dance and a culture that affirms everyone's differences and celebrates dance as a means of expression.

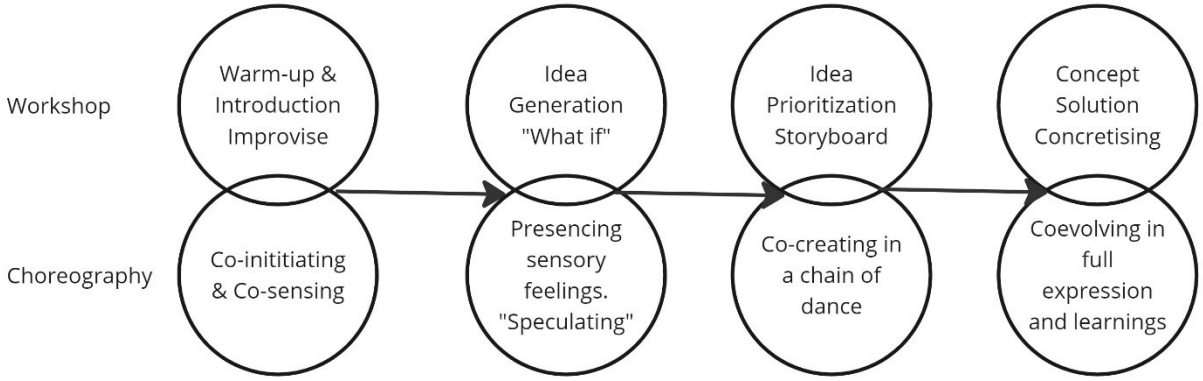


Figure 31: Comparing the workshop process with the Choreography process (Scharmer, 2007).

Co-sensing: observe, listen with mind and heart open, using all senses without habitual judgement (Scharmer, 2007): I had not attended a party in many years, I was nervous about being center of attention and still feel excluded. I was surprised/happy with the arrangement, spectators formed a circle around the choreographers, there was no hierarchy, just focused on the activity in the middle. I noticed the diversity of the choreographers and spectators. I noticed the facilitator on the piano, and the DJ playing.

Presencing: connect to the source of inspiration (Scharmer, 2007): The spectator's energy was transformed with each variation of the beat, piano, lights, smoke effects, the movement of the choreographers as the facilitator (acting as instrumentalist, dancer, facilitator) guided them through different speculative scenarios of movement. The spectators nodded/ tapped to the beat, improvised along with the dancers in the circular position.

Co-creating prototype the new- and explore the future by doing, use of senses (Scharmer, 2007): The spectators were prompted to move round in a chain, led by the facilitator, improvising movement.

Coevolving: embody the new in ecosystems- that facilitate seeing and acting from the whole(Scharmer, 2007). And finally, the climax of loud music and people joyfully dancing along. Through shared emotional states, there were possibilities for new mutual relationships. I realised that choreography could be used as a point of entry for food related activities in the neighbourhood. Though choreography is a very intentional design, it is without the niceties/etiquette associated with food, such as - dressing right, using the right plate/cutlery, dishing the food right, food having the right temperature, right dining ambiance, right conversation, right people; all these can paralyze the thinking of organisers and visitors at community events geared towards inclusion.

The Choreography and the workshop design process created synergy. Through collective use, each individual participant gets in touch with emotions (fear, anxiety, joy), becomes conscious of emotional states, and creates new states that are advantageous for the group's situated action. A social flow, including both individuals who like solitude and collective activities in co-creative settings, alone but together (Zenk et al., 2021), has great possibilities for neighbourhood food initiatives.

Discussion on Solution phase- Creating scenarios, artifacts and solutions in the workshop was an iterative process of creating new solutions and testing them. The solution activity is one of exploring, experimentation and learning. The learning style suited for creating solutions is the accommodation style. This is for the purpose of designing an alternative solution to a sustainable shared kitchen environment. Learnings from the workshop and choreography: Integrated autonomy involves the degree of freedom the participants can act in relation to the social structures (Backström, 2013). At the workshop, integrated autonomy was experienced during the warmup stage. In the choreography, as the dancers were performing, the spectators were individually listening, reflecting and nodding along.

Convergent divergence means information gathering and the consequences of the choices made (Backström, 2013). At the workshop, participants were involved in idea generation aimed at including people in activities, they converged ideas using a storyboard and concretised how the idea would work. At the choreography, with the prompting of the facilitator the spectators danced round in a circle, then afterwards people danced to the music, experiencing and expressing themselves and forming new acquaintances/meanings. As I participated in the dance, I was inspired to include choreography in the kitchen design concept which is in line with research by Österberg & Olsson (2021) that brief periods of dancing can increase willingness to engage, explorative thinking of idea generation and reduce functional fixedness. Connecting the processes of the dance to my thesis shows my level of mindfulness, curiosity, openness and acceptance.

4.7 Scenario Design

The solution stage highlighted the activities in the workshop and compared with the activities in the choreography. The findings from the workshop shows that a common understanding of planning a social event based on people's common interests is of value to the residents. Also, using the thesis model that I created in the framework section, the findings in the workshop suggests that family welfare value can be experienced within homes, across multi-networks and within neighbourhoods - when occupants cocreate around common interests, enabled by different user and organisational resources such as: smart manufacturing, AI/technology in determining price and ingredients, Digital Twin, change in behaviour towards food wastage and reuse; and connecting climate change with mortality to improve health and well-being.

Scenario design: I combined the findings from the workshop and the ethnography research to generate conceptual scenarios. Looking back at the design codes in the framework stage. Considerations based on the user needs (for security, privacy, socializing, wellbeing, exercise, healthy food, connection to food source, sustainable living, autonomy) and political structures (control, bureaucracy, effect on behaviour, social inequalities, dominant groups, environmental hazards); the kitchen concepts (food production, upcycling waste, good eating habits, dynamic space, social interaction and functionality) gave insights to possible scenarios. I identified interesting stories from the data and grouped them based on food wastage, circularity, technology and social (Table 15).

Food Wastage	Circularity	Agriculture/Tech	Social
Influencing the use of products Garden compost. Anaerobic digestor Recycle waste within buildings. Sustainable kitchens Improve storage. Recycling machine	Biomass Renewable energy for power BIG Floating city Fresh water autonomy – roof collection	Precedents Aquaponics, indoor farming Functionality Dynamic kitchen Garden in kitchen ReGen Villages	Interactive Multi-sensory activity Training for work Connection to food source Stereotype affecting design. Food initiatives Political structures More motivations Choreography + art Common interests

Table 15: Interesting stories in ethnography and workshop

Using the dimensions for innovation such as- offerings, customers, processes and presence (Sawhney et al. 2006), I designed scenarios based on the dimensions of food wastage, circularity, Agriculture/technology and social (Table 16).

Innovation Dimension	Dimension	Sources	Speculation
Processes & New Networks Redesign core operating processes	Food Waste & Circularity & Technology	Questionnaire result Interview with Participant M Precedents Workshop	What if food waste can be extracted directly from the kitchen for other use; and the required food is directly supplied to the kitchen.
Offerings & New Networks Develop an innovative new service	Agriculture & Technology	Focus Group Precedents Workshop Questionnaire	What if we were all farmers, and our occupation is directly related to food production and nature care.
Customers & New Networks Discover unmet occupants needs	Social	Observations Workshop Prototype of food event Interview Questionnaire	What if choreography can connect

Table 16: Innovation dimensions and Speculations

Food waste, Circularity and Technology: In the questionnaire response, on how professionals can reduce wastage, one of the participants (Construction PM) highlighted the use of "Innovative domesticated technologies that recycle waste within buildings". This concept also captures other ideas mentioned in the questionnaire, interview, precedents and the workshop.

Agriculture & Technology: In the focus group, one of the participants questioned the food source and how it can be integrated into a kitchen? She questioned the relationship of people and food; and the connection to health. The idea of food and architecture/urban was also

explored in the precedents; the workshop and questionnaire also mentioned how technology can enable functionality of the food system/kitchen.

Social: My experience and observation during the choreography brought about the idea of dance movements as an entry point towards resident-driven food initiatives. Project managing a food event, interviews and questionnaires highlighted the need for different motivations for social interactions.

Dystopia perspective: this includes - none addressed political structures of control, bureaucracy, effect on behaviour, social inequalities, environmental hazards. Also lack of mutual understanding, not addressing different motivations and lack of leadership that can harness the diverse, complex and ever dynamic landscape of the neighbourhood.

In order to illustrate the systemic problems and opportunities of an alternative shared kitchen, scenarios were developed. This illustrates users' experiences in the neighbourhood, with food and the kitchen.

1. What if we can bypass supermarkets and the refuse dump room; and the food waste can be extracted directly from the kitchen for other use; and the required food is directly supplied to the kitchen.

> AI technologies are used to predict timely supply of healthy food required in kitchens and personalised based on health.

2. What if choreography can connect the unconnected people in the neighbourhood.

> Social dance and music liberating lack of prestige as the point of entry to alternative shared kitchen experiences in intercultural neighbourhoods.

3. What if we were all farmers, and our occupation is directly related to food production and nature care.

> Distance between professions and regenerative care for land and nature is reduced.

4. What if our gene to be empathetic becomes redundant, because we no longer use the emotion. Continued lack of active listening, agreeableness, emotional intelligence; continued lack of eye contact, physical and psychological distance- causing systemic dehumanization.

> Babies are produced on earth and cared for by robots, while adults live on Mars-controlled unknowingly by robots. People are not tied to the place/land of birth and can identify with any country/language/culture/gender they desire.

4.7.1 Design Concept

This thesis attempts to visualize the scenario of a self-sustaining food network that is centered on resident-driven initiatives creating social networks, healthy eating and well-being (Figure 32). A dystopia perspective is continued unhealthy and segregated living.

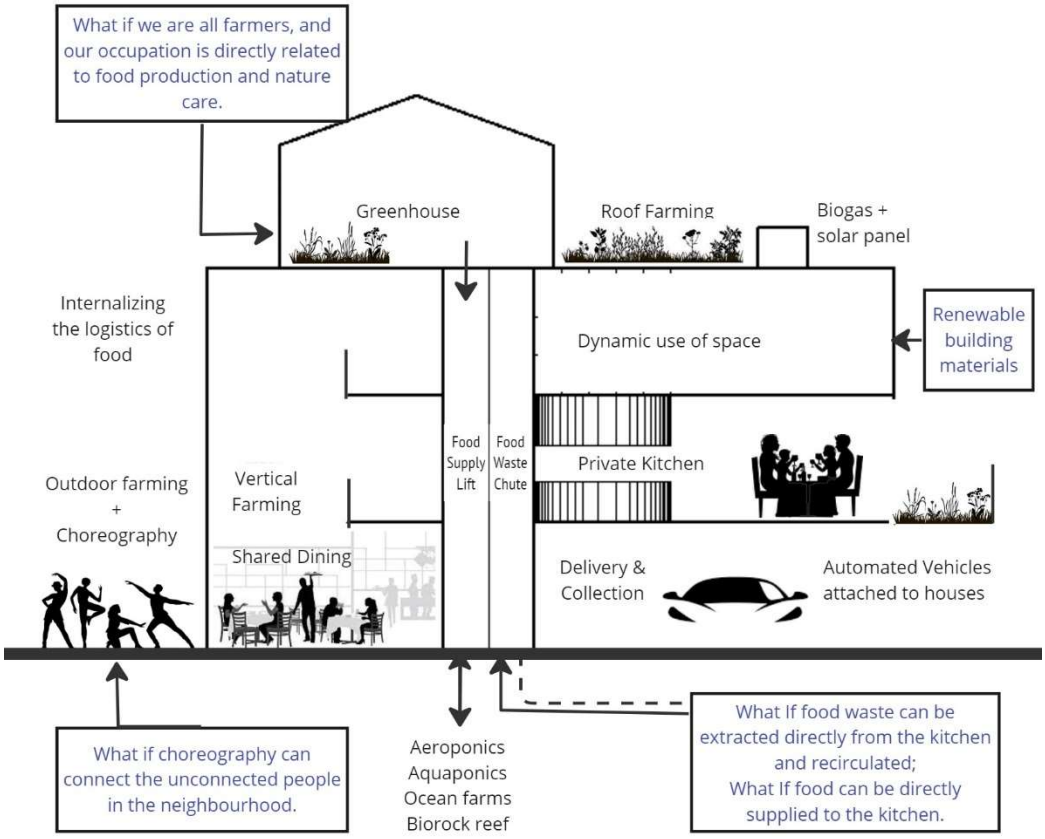


Figure 32: Design scenarios

4.8 Summary of the Innovation Process

4.8.1 Summary of the Results

Looking at the innovation process as one of problem finding, problem selecting, solution finding and solution selecting. *The ethnography studies* (problem finding): I was able to identify the need for an alternative sustainable social kitchen. One of the problems that I have been trying to address in this thesis is to balance the social aspect of sustainability with the other aspects of ecology and economy. Also mentioned by Hele Landet in Thinking outside the B(l)ox, many strategies and plans have attempted to create a framework for good life across generations however the strategies had no effects and communities still face physical, social and organizational challenges. Understanding the core needs is important. The interviews, focus group, photo studies and storyboarding used in the ethnography phase

identified some problems and needs such as need for regenerative spaces that are dynamic and flexible, the need to know the food source, social interaction and healthy eating/ living for the well-being of people. Different social structures and boundaries were also identified in the ethnography study.

The framework stage (problem selection): I extracted important insight and looked for gaps. The interesting new stories that I had to focus on were the farm, healthy eating, and a dynamic space. My initial focus was on technology as a means for an alternative kitchen however, new stories, new connections shifted the focus. This resulted in rearranging the themes/patterns and coming up with a simple model. I drew conclusions at this phase that, with the family welfare positioned at the center, they have the ability for sustainable wellbeing when connected to a sustainable shared kitchen with relevant parties of food, health, urban neighbourhood, government and public sectors. *The imperative stage*, also seen as (solution finding)- this involved planning a workshop, and designing the workshop based on design principles that should yield responsible innovation. *The solution stage*- (solution selecting) was done using artifacts and storyboards in producing concept solutions. This was tested using a concretization tool- showing resources required to actualize the solution.

5 Theoretical Lessons Learned

This study seeks to understand how systemic, regenerative, and speculative design be integrated to address the challenges in shared kitchen environments. The study also aims to understand the needs and preferences of diverse user groups in shared kitchen spaces. To this end, analysis/ learnings from the workshop, ethnography study, regenerative sustainable design, Systemic design; and social /cultural critique are detailed below.

5.1 Regenerative Sustainability Design

L. V. Gibbons (2020) mentioned that embedded throughout the process and product domains of development and design initiatives, regenerative development principles and core attributes are integrated across the ecological and sociocultural aspects of living systems (Table 17).

Category Shared Kitchen Design Concept	Regenerative Community Development Strategies
Guiding Consciousness The Kitchen can be seen to be interconnected to other functions and ecosystems such as health sector, food production, transportation, technologies, urban etc. It is multifunctional and dynamic that the kitchen space could be used to meet diverse user needs- healthy eating habits, social space and integrates ecological and social dimensions.	Design of systems Developmental processes, goals, outcomes Metadesign (shifts worldviews) Ecological design, integrated ecologies Conscious and intentional actions Implementing indigenous practices
Actions This thesis involved a lot of participatory methods involving people in focus groups, workshops and other ethnography studies to integrate multiple perspectives of understanding people's kitchen needs and improving the kitchen for the future.	Integrating multiple perspectives Co-creativity Deep participation and dialogue Designed experiments, adaptive design
Community Building Culture This thesis will be pointless without the emphasis on the importance of the equity of the social, environmental and economic aspect of sustainability. As mentioned in the focus group, the kitchen has become historically and socially dysfunctional and there is the need to connect the agricultural and cultural diversity of the community for vitality and life. This also includes educating people about regenerative practices and vision for sustainability.	Celebrations based around healthy living system functioning. Equity (social, ecological) Inclusivity Local economies Community contributions of time/efforts/material goods Satisfying or Purposeful livelihoods Increasing human health, well-being, happiness
Governance The community will need to be carried along in policies, leaving no one behind (e.g persona "Different"), in regenerative development. CO2 reporting.	Transparency Accountability Long-term and short-term view

Health Increase care for places, educate the community about food wastage, energy consumption, healthy eating, social inclusion. Long life and prosperity, communal kitchen that adapts to food scarcities and reduces loneliness.	Increasing human health, well-being Increased ecological health
Circularity: Use biodegradable materials for kitchen building. Agriculture: Food production/ towers integrated into building design. Waste: use of rainwater and greywater, treat and extract nutrients nitrogen and phosphorus from black water before it flows offsite. Waste sorting integrated into building design. Technology: green logistics, smart manufacturing, AI for recipes/spices/health checks. Autonomous Vehicles connected to home. Personalised experiences, flexible and multi-layered solutions in the kitchen, decision making.	

Table 17: Regenerative kitchen analysis (L. V. Gibbons, 2020)

Value Creation in Regenerative Design

An organisation with a regenerative business model proposes, generates, and distributes value in harmony with the environment, while also facilitating value capture at every stakeholder level. Elements of value proposition, value delivery and creation, and value capture (Konietzko et al., 2023) are detailed below in (Table 18).

Value	Value created
Value Proposition Planetary health and societal well-being	A holistic value proposition that prioritizes physical, mental and social wellbeing of societies, customers, suppliers, employees and tenants using the kitchen and food networks. Using plant based, biodegradable materials, sustainable packaging, ethical business practices and enhanced well-being of people.
Value creation and Delivery Regenerative Leadership	Intentional focus on minimizing environmental footprint, addressing social challenges, demonstrating transparency and commitment to health and welfare, clear mission on holistic wellbeing.
Co-creative partnership with nature	Nature-inclusive design; climate resilience e.g passive cooling, thermal comfort, reducing energy consumption; Natural capital investment; connection to nature through activities on team building, volunteering; Nature-based solutions to solve climate change, food and water security; use of indigenous knowledge.
Justice and fairness	Nature and animal right advocacy. Fair labor practices, ethical sourcing, diversity and inclusion, addressing systemic discrimination; responsible sourcing; balance social, environmental impact and financial returns.
Value Capture Multi-capital accounting and net positive impact	Assessment of environmental impact, consumption practices, social cost, financial capital; prioritize employee development, well-being and support. For example internships and jobs provided in the social kitchen

Table 18: Value-added Kitchen (Konietzko et al., 2023)

5.2 Systemic Design

Common elements that an organisation should take into account in order to achieve kitchen-food innovation effectively, are revealed through a review of innovation management frameworks using systems approach (Karlsson & Magnusson, 2019). Each of the following eight themes can be applied to these components: context, direction, leadership, culture, processes, structures, support and resources, and evaluation (Table 19).

Theme	Type of Elements
Context	Trends: AI, ChatGPT, smart manufacturing, Digital Twin Opportunities: from precedents studied, we identified opportunities in neighbourhood design for collaborative food and kitchen activities. User needs and requirements: need for safety and security, healthy and comfortable lifestyle, Stakeholders: homeowners, tenants, property developers, housing associations, neighbourhood, farmers, suppliers, logistics,
Direction	Vision: this ensures that the relevant people know the vision, strategies, policies and purpose of having an alternative shared kitchen. Ensuring that the kitchen can contribute to improved health and welfare, reduce wastage and improve healthy eating habits
Leadership	Leadership styles and values Realising value, exploiting insights, future-focused leaders, systems approach, adaptable structures, innovation culture, strategic direction, managing uncertainty. This requires leaders to plan -act -check and execute the vision in an iterative manner.
Culture	Social context, culture supporting innovation activities
Processes	Innovation process: this includes identifying opportunities and needs in the ethnography studies, creating concepts and scenarios in the workshop for inviting people for food activities; we saw the need to develop a kitchen solution for circularity, reduce wastage, increase social connections. The solution is expected to connect the unconnected people and nature.
Structures	Organizational set up, will ensure proper communications of what, why, when, how and who - related to different stakeholders (customers, partners, suppliers, property developers etc)
Support and resources	Ensuring that the intellectual property rights of the parties involved in intercultural knowledge of indigenous/modern technologies of building sustainable kitchens and food making is protected; ensure resources (people, time, knowledge, funds) are available at the right time.
Evaluation	Indicator metrics

Table 19: Systemic perspective (Karlsson & Magnusson, 2019)

Leadership: In designing a shared alternative kitchen that is more social, there is a paradoxical need to keep/protect the authentic culture of a place/people and the need to include the diversity of the neighbourhood. The type of leadership of the different stakeholder organizations such as property developers, architects, housing associations, public sectors, and tenants will require dynamic capabilities such as deliberate process for idea management, building a creative climate and an inclusive approach to leadership. They must have the

capacity to lead change from top-down and bottom-up perspectives and have a change strategy that allows for emergent influences (Isaksen & Tidd, 2006).

According to Backström, T. (2017) a dynamic organization is an organization that is ambidextrous and can manage to combine the dichotomy of doing things right (exploitation, stability, control, and efficiency) and doing the right things (exploration, adaptability, creativity, and effectiveness). Tenants require basic psychological needs/ motivation for healthy developments such as autonomy, competence, a sense of belonging and connection (Ryan & Deci, 2020). Successful leaders should be able to challenge processes and try new ways, inspire, involve others to act, set examples to follow and recognise peoples' contributions. For example, participative management style and vision have the strongest influence on team dynamics (Sarin & O'Connor, 2009). That means that landlords will need more participatory methods to motivate tenants to food/ kitchen related activities. According to (Wikström, 2022) strategy and tactics enables clarity in innovative work (i.e of a shared kitchen); tactics and people enable control using skills and capability; strategy and people improve self confidence of tenants to participate, self organise themselves and contribute meaningfully.

5.3 Ethnography Study

The data collected in the ethnography study was affected by my senses, language, pre-understanding, intentions and theories (as a private, secure person). I drew conclusions about social structures and the dynamics of the system. From the observations of student housing apartments, I identified how the architect's designs can affect the social interaction of a place. There is a level of control in the distribution of type, number, status of people in neighbourhoods by housing marketplace. In the precedent, I also identified some innovative food groups for socialization such as neighbourhood gardens, food banks for community events, eating together at religious events, spreading awareness on sustainability, volunteering on advocating against food waste, food and film clubs, nature detectives. All these are excellent initiatives, however if there is no tolerance towards openness and inclusion, we would have a repeat of the current situation of people avoiding neighbours. Also, during the interviews and focus groups conducted, my direct questions regarding if people were willing to share a space resulted in latent answers of being associated with someone different. This was helpful for the thesis. I had organised a food event previously and witnessed the diverse tensions related to quality of food and type of event. Cultural differences of practice lead to negotiation of different identities, which do not harmoniously coexist (Akkerman & Bakker, 2011). Intercultural events can have tensions that reflect structural conflicts such as identity of the organizer, whose idea is being championed, whose narrative is promoted, to what extent is a cultural narrative promoted amongst the diversity of other cultures, what will history say of the achievement? These are some questions that could prevent integration in an alternative shared kitchen and promote segregated silos of achievement. According to (Hülsheger et al., 2009) job-relevant diversity (displaying vision, task orientation, external communication, support for innovation) seem more conducive to

innovation than background diversity at the workplace. This could be one of the reasons for lack of motivation for background diversity in some communities. Hoever et al. (2012) proposed perspective taking to realize the creative benefits of diversity of perspectives by fostering information elaboration.

5.4 Social and Cultural Critique

> "A man who calls his kinsmen to a feast does not do so to save them from starving. They all have food in their own homes. When we gather together in the moonlit village ground it is not because of the moon. Every man can see it in his own compound. We come together because it is good for kinsmen to do so." Chinua Achebe's Things Fall Apart- 1958.

The emphasis that I would like to focus on Chinua Achebe's quote above, is that we can all be individualistic, but we need to decide individually and collectively to work together in harmony. Inclusion should go beyond kinsmen, pets, preschool friends, nature, classmates, common cultural countries, non-human; and be open to discover new commons in people different from us.

In the warming climate [.. it] reminded us not to follow the stereotypical thinking of 'conventional wisdom' but to re-think each design decision clearly for ourselves and remind us that the sustainability stool has three legs, and if you lose one, the stool falls (Roaf, 2008).

Modern innovation and design practice: In today's world, Design thinking, participatory design and responsible research innovation is very important, and designers/ researchers should be accountable. Still relevant are Victor Pananek's ideas focused on social design and design thinking for "Different" (Appendix 3) using sustainable designs (Papanek et al., 2010). Responsible innovation is not just about producing usable innovations; it involves the ways and manner we interact, empathize, co-create, implement and follow-up (Wikberg Nilsson, 2021) with the clients and stakeholders and the processes involved. I will even say, do not start a project if it will dehumanize people. Embedded in the late 1960s pan-Scandinavian design politics is the requirement for socially responsible design (Clarke, 2013) and participatory designs (Binder et al., 2011). Moreover, responsible use of artificial intelligence such as ChatGPT will be required to avoid issues such as muted narratives, devaluation of relationships, unemployment, privacy concerns, bias, misinformation and digital inequities (Polyportis & Pahos, 2024). Buchanan (2001) recommended us to realise that our communications and constructions may be actions; and that the designer should be accountable while designing for a diversity of people.

Situated knowledge necessitates the representation of the object of knowledge as an active participant and agent, rather than a mere resource, screen, or ground, and never as a subordinate to the master whose exclusive authority and authorship of "objective" knowledge stifle the dialectic (Haraway, 2022). Establishing a shift in the "distribution of the sensible" by substituting situatedness for context as the defining term of engagement enables architects to create sites of intervention through situated matters of concern and care, thereby fostering

a profound ecological mode of operation. The architect uses situatedness to choreograph thinking and creation, thereby exploring the capacity of architecture to foster empathic connections among people, materials, animals, processes, and concerns (Lieberman, 2019).

According to Smith (2016), in complex projects involving the collaboration of individuals from various organisations and disciplines, such as a shared kitchen neighbourhood, there is a tendency for individuals to react to uncertainty and try to reduce it. They have the ability to act when the individual or group has effectively understood the processes of the shared neighbourhood kitchen. However, in the event that an individual or group is unable to understand the process, they disengage from the collaborative endeavour. Implementing goal-oriented strategies to reduce uncertainties can result in preventing the opportunities of others, thereby fostering mistrust and diminishing interpersonal engagements. Self-esteem, security, and a sense of belonging are all benefits of group identity. As a result, organisations become segregated and fail to interact with one another. Thus, in a neighbourhood network, project managers must coordinate the efforts of individuals utilising sensemaking, group identification, and strategizing to reduce uncertainty. It is crucial that they remain mindful of the fundamental mechanisms that govern the formation of subgroups along knowledge boundaries and prevent them from becoming hermetic and fostering exclusion (Smith, 2016).

According to Leonie Sandercock (2003) the concept of "community" could lose its relevant in today's society, "in which difference, otherness, fragmentation, splintering, multiplicity, heterogeneity, diversity, (and) plurality prevail", and where social ties are predominantly virtual. Many people live in cities with different neighbours. People exclude themselves from social initiatives because they don’t feel they belong in public spaces that should be organically social. Shared tasks enable temporary freedom from restrictive social systems and foster a feeling of common humanity. Guevara et al. (2023) stated that students will be more capable of dismantling stereotypes of determinism and essentialism if an undergraduate curriculum emphasises human evolutionary history, complex trait architecture, and the global distribution of human genetic variation.

The behaviour of complex organisations is nonlinear. Establishing and modifying environments that foster the emergence of effective, self-organized, improvised solutions is the strategic direction of complex organisations, as determined by the tenants. By reconfiguring the organisational architecture in which agents adapt and modifying the fitness landscape for local agents, managers exert an influence on strategic behaviour (Anderson, 1999). Furthermore, within the framework of ongoing strategy development, strategic communication serves as a means to oversee the integration of communication processes. This encompasses not only the continuous loop of building and rebuilding of strategies, but also their presentation, promotion, and execution (van Ruler, 2018).

Choreography and food events can be seen as boundary crossing where people enter into a territory unfamiliar and are faced with the challenge of negotiating and combining elements from different contexts to achieve hybrid solutions. Boundary crossing refers to the process of establishing continuity amidst sociocultural differences, in which the exchange of established practices yields novel insights by virtue of their differences (Akkerman & Bakker,

2011). Karl Linn (1923 -2005) believed temporary “staging” of choreographed means of participation and collective engagement could evolve into a more permanent physical framework if the rituals of co-designing with users in workshops proved resilient (Hirsch, 2015).

Developers, public sectors and housing associations will need to increase the success of improving social networks by practicing skills in learning, leading, relating and cycling the divergent and convergent activities through the innovation process (Van de Ven, 2017). In solving sustainability issues, focus will be needed in creating conditions by which social innovation and entrepreneurship will flourish. Schumpeter defines the socialist society as an institutional structure in which the society, rather than the private sector, possesses ownership. Social value is the result of all members of society acting collectively and consciously as a community under the guidance of an enterprising, vigorous, and ambidextrous leadership style (Schumpeter, 2017). According to Oeij et al. (2019) elements leading to the adoption of social innovation include power-based design, powerful people and leadership, and incremental progress.

6 Discussion

Critical realism was used to evaluate the qualitative method including ethnography research and workshop activities, and phenomenological methods used in this thesis. Data was gotten abductively from literature studies and qualitative methods in trying to understand the needs of people using the kitchen. I recontextualized the groupings and reinterpreted the output of the data, drawing conclusions from the final phase of the ethnography studies which was the focus group discussion. The workshop activities unearth more latent needs from the participants. The workshop was created in an Enabling Space where processes for creating new knowledge and innovation can emerge. I ensured to practice an enabling attitude for facilitating, observing, and participating in the workshop where; - physical, social, cognitive, technological, epistemological, cultural, intellectual, emotional and other factors; were considered and integrated in the creative problem-solving process.

The goal of the thesis project is to explore to understand an alternative social kitchen and stimulate a shift towards more inclusive and environmentally conscious design practices. This was done by exploring the potential of systemic, regenerative, speculative sustainable design in addressing challenges faced in shared kitchens, while understanding diverse user needs, motivations and expectations within enabling contexts. I collected relevant information from the ethnography studies and designed a workshop that stimulated participants in generating sustainable scenarios using storyboard tools in a speculative approach. With the social kitchen hub at the center, different professionals addressed the improvement of the kitchen from different contexts of functionality, food wastage, social interaction and circularity. The research analysis revealed diverse needs of users such as social connection, more functional/dynamic/flexible kitchen space, knowledge about food source and recipes, and healthy eating habits. The regenerative design analysis showed design concepts, community design strategies of the kitchen and also value propositions at different stakeholder involvement. The evaluation helps people work holistically with complex living systems exploring key life-giving elements and their relationships across scales (L. V. Gibbons, 2020; Konietzko et al., 2023). The systemic design identified the roles leadership play in the people, processes, product and place (Isaksen & Tidd, 2006; Tidd & Bessant, 2020). The speculative design of possible near future worlds highlighted how different artifacts were used to communicate with users and elicit different ideas and needs (Dunne & Raby, 2013). A critical look at the social aspect of the shared kitchen was explored and I highlighted the role project managers/leaders will have to play in orchestrating and harnessing the complexity of shared neighbourhoods. If we are to take seriously the proposals of shared activities in neighbourhoods, we will need individual mind shifts and leadership that has the emotional intelligence, boundary knowledge, empathy, vision, ability to enable-coach and steer diversity of the community for vitality and life. Choreography and other forms of arts was proposed as a ritual that could bridge the gap and reduce generational stereotypical behaviours.

Social Practice

While we have organisations and people promoting diversity and inclusion / internationalization, there are some who have the nostalgic feeling of when Europe was still Europe, or when there were no 'black and brown' people. Despite my bold workshop of introducing the persona "Different", and despite highlighting dominant group behaviours that could occur during food event initiatives; The need for social inclusion in shared kitchen neighbourhoods could still be viewed as an individual problem only. 'Different' will be required to acclimatise into the society by gaining more social experience or move location.

According to Hernes et al. (2013) the continuous flow of present actions that concurrently draw from the past and the future as epistemic resources—which are themselves susceptible to perpetual reconstruction—is how temporal continuity is articulated. People arrive at shared understandings of reality through temporal experience within the intersubjective dimensions of sociality. Process thinkers like Whitehead, Mead, Heidegger, and Ricoeur advocate for a non-intentional teleology, where futures and pasts co-constitute. Understanding the emergent nature of the present enhances knowledge of how the present connects to the past while shaping the future (Hernes et al., 2013). With a systemic perspective of looking at historic intercultural design attitudes, I concluded that an alternative shared kitchen is not an individual problem but a collective problem to be solved by all.

Mirroring behaviour

Relating conformity to the city, I would say newcomers in the neighbourhood mirror the attitude of the settled/host community resulting in a constellation of mirrored attitudes/ behaviours (good and bad). Also, the distance between the host/leadership and newcomer/ followers affects the level of influence. According to (Olsson & Florin, 2011), groups that understand improvisation are better equipped to handle unfavourable outcomes like conformity, pressure and groupthink.

Zhou & Luo (2012) highlighted threats to group interactions as: social laziness, conformity, stifling of production, and establishment of downward norms. In a knowledge creation spiral of tacit to explicit knowledge, people experience socialization, externalization, combination and internalization. According to (Meindl, 1995) social contagion view of leadership emphasises how ideas about leadership gradually diffuse throughout a group of followers. Group composition affects follower interaction dynamics. Heterogeneous groups increase the diversity of perspectives, attitudes, and opinions on task-related subjects and change the socioemotional climate through situated identities and stereotypes based on prominent social categories. The resulting interaction dynamics have the potential to shape leadership by influencing cognitive and affective attitudes. Zhou & Luo (2012) mentioned that as understanding advances, ideas are discussed, contested, appropriated, and disseminated through collaborative engagement. Achieving such comprehension is a fundamentally collaborative and social process, constituting a dialogical phenomenon.

Empathy

In my opinion, without empathy it is not possible to produce the best designs /innovations. We could try using questionnaires or AI to understand people but when compared with actual empathy motivated designs, the difference is clear. According to (Decety & Jackson, 2004) empathy is a motivated process that is triggered voluntarily and is a method of gaining knowledge of understanding another. An inclusive design requires putting people at the center of the design-making process and enabling embodied experiences (Luck, 2018). Eggen et al. (2016) and Hoover (2018) recommended that designers should analyse and understand the ecosystem of where they will be working (having a holistic user experience paradigm); build effective partnerships; develop a solution that will actually be adopted; ensure the solution improves the lives of people they are design for; develop a business model that is self sustaining and integrated in everyday life.

7 Conclusion

This thesis project proposes an alternative shared kitchen space in the role of challenging status-quo of the current kitchen environment by using systemic, regenerative and speculative design to address problems in housing communities. The thesis presented an innovation and learning process of four stages: Ethnography research, Framework, Imperatives and Solution. This process supports exploring to understand an alternative shared kitchen that is people-driven around shared interests, promoting social well-being. The innovation process involved moving from ethnography study to frameworks to solutions and back to frameworks to form meaningful imperatives of the process. This unearths the needs and preferences of diverse people in the shared kitchen and also reveals structures and mechanisms. This is to visualize new conceptions and critical imaginaries that can cause systemic change. Diverse learning styles of experiencing, reflecting, thinking and acting was explored throughout the process.

The approach brought an understanding of user needs in shared kitchen spaces with a systemic, regenerative, and speculative design perspective. It observed the user needs (for security, privacy, socializing, wellbeing, exercise, healthy food, connection to food source, sustainable living, autonomy) and political structures (control, bureaucracy, effect on behaviour, social inequalities, dominant groups, environmental hazards); the kitchen concepts (food production, upcycling waste, good eating habits, dynamic space, social interaction and functionality). This provided insights into the workshop and speculative designs. In this respect, the result suggests that when residents cocreate around shared interests, family welfare value can be experienced in homes, across multiple networks, and within neighbourhoods. The welfare of people in the kitchen hub is made possible by a variety of user and organisational resources, such as social activities, sustainable food production, sustainable food waste collection, technology, waste recirculation and housing renewable energy, for a holistic health and wellbeing in neighbourhoods. The thesis project illustrates how the kitchen environment can support social interactions, good food, health and lifestyle and wellbeing of the society; and be resilient to future uncertainties.

7.1 Contribution to the study

Prioritising social well-being and loneliness requires participation/ involvement in communities, belonging, trust in others and safety. Citizens-driven initiatives makes it easy to join communities, Resident-driven initiatives around shared interests can bridge ethnic and cultural divides in neighbourhoods such as cooking pavilion, choreography, growing gardens together, art making, while ensuring openness as to who can participate in the co-creation. Creating spaces that can easily be reprogrammed for temporary use and self-organised activities. The use of urban nature and physical activity can help people exercise, relax and socialize across social and cultural divides (Juul, 2023). This thesis research could open dialogue from designers and non-designers of communities, researchers, politicians, urban planners, architects, housing associations and the public sector for a holistic, cross-disciplinary focus across urban development, sustainability and well-being.

7.2 Limitations and future research

In order to explore concept solutions based on core user needs for an alternative shared kitchen, a workshop was organised to gain perspectives from different professionals. Only four participants were involved in the workshop. Further research could include more participants for more innovative ideas and prototype solutions for kitchen designs in neighbourhoods. A systemic approach of an alternative shared kitchen was explored in this thesis, a closer look at the functionality of kitchen spatial design for promoting social interactions can be investigated. E.g the kitchen island as a dynamic, flexible and social space. My observation of neighbourhood housing was from my perspective as an international student living in a private housing unit for two years. Locals with strong family and social networks may have a different experience in their neighbourhood. My personality as somewhat of an individualistic-social and an introvert-extrovert may have affected choosing design solutions for a direct food supply and waste disposal. The implication is that people will spend more time indoors. Further research could be looked into outdoor activities for neighbourhood design or open housing typologies. This thesis took a brief look at the ethnic minority perspective for an inclusive design, other marginalized groups for social inclusion such as persons with disabilities, unemployed youth, gender and sexual minorities, elderly, children, people with limited education could be looked at. More first-persons/ marginalised / muted narratives could be researched. Also, housing landlords were not interviewed in this thesis, it will be interesting to know requests made by residents, to understand their latent needs e.g not wanting/wanting to be in the same block as a schoolmate, workmate or family with children. Finally, due to different interests and personalities, it is not possible to integrate everyone into neighbourhood activities, further research into new meanings of working together but alone, could be explored.

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Appendix

Appendix 0: Innovation and Design Management - Course Contribution

Innovation and Creativity management was the starting point. The course's workshop session had built my confidence as a facilitator to focus on creating an Enabling Space. The innovation process, learning process, creativity and systemic thinking was adequately captured this thesis. While exploring ideas in this thesis - I was able to recognise the need to include relevant data from other courses such as the outcome of the food event of *Project management course*; and gather ethnography research in the *Early phases in innovation and design*- and the ethics involved; As the solution finding became more complex, I included the perspective of boundary crossing from *Communication in complex organisations*; and by delving deeper into solution finding- I made use of situated design as a Research method; and because the problem of a shared kitchen seemed to be an individual responsibility, I went back to the ethnography research and uncovered an historical perspective highlighting the *Challenges in innovation and design*. The discovery of political structures and stereotypical behaviours highlighted a collective/ shared responsibility towards inclusion. The stereotypical behaviour is not specific to a country/location as exclusion is everywhere. Design thinking, participatory design, responsible research innovation and Co-creation from *Human Centered Design* perspective was recommended for addressing an alternative shared kitchen.

Appendix 1: Questionnaire (inspired by neighbourhoodsforgenerations.com/en/winner/feed-back). I added the professional perspective.

1. Gender?
2. Age?
3. Profession/Course of Study/Career interest?
4. Household type?
5. What don't you like about the kitchen?
6. How do you imagine the kitchen in the future?
7. Could preparing meals become a social activity across generations, mitigating segregation?
8. How can your profession/ personal interest (e.g as a farmer, IT expert, artist...) make it easier to use the kitchen for social interactions in the future?
9. How can you enhance the functionality of the kitchen?
10. What do you not eat that you wish you did?
11. How can your profession contribute to encouraging healthy eating habits?
12. Could our neighbourhoods start producing their own food and upcycle waste?
13. What do you throw away that disturbs you?
14. How can your profession/career interest (e.g as a farmer, IT expert) make it easier to reduce wastage in the future kitchen?
15. What can you change in the kitchen and make in new ways?
16. Where do you get your recipes?

How do you imagine the kitchen in the future?

11 responses

Pretty much like we have seen today, maybe with more smart convertable shelves, cabinets, hidden cabinets in the walls to have more spaces.

Well ventilation, offering adequate thermal comfort within

A lively place where social activities can take place.

To be private, fully equipped and a bit cozy

Powered by AI

To be spacious and properly ventilated

Where food will cook its self

Smart

More modernize

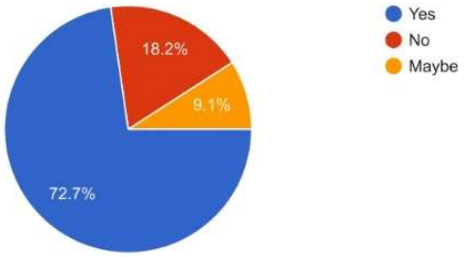
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Could preparing meals become a social activity across generations, mitigating segregation?

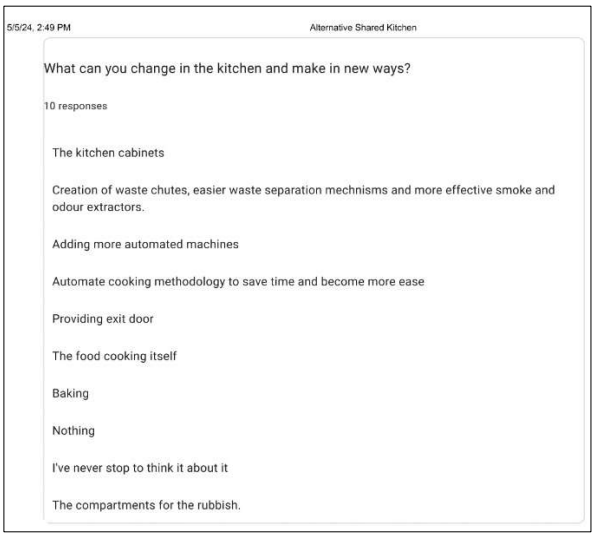
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11 responses

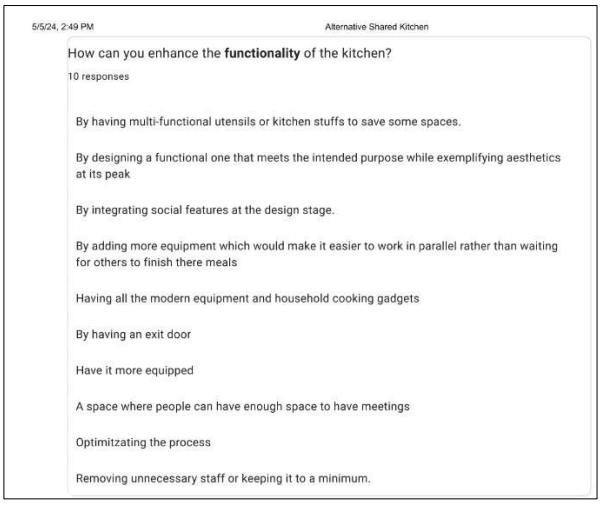



5/5/24, 2:49 PM
Alternative Shared Kitchen
What do you not eat that you wish you did?
9 responses
Fish, I prefer not to cook any because the smell stays for some time.
Pounded yam
Sea food
Nothing
Durian
Nothing
More African meals
Big steaks
Fresh fish


5/5/24, 2:49 PM
Alternative Shared Kitchen
What do you throw away that disturbs you?
10 responses
Old-scratched-pans
Water
Food items
Leftovers
Food remnants
Bread
Things used to cook and remaining old food
Food because they expire early
I wouldn't throw it if didn't want to
The washing machine

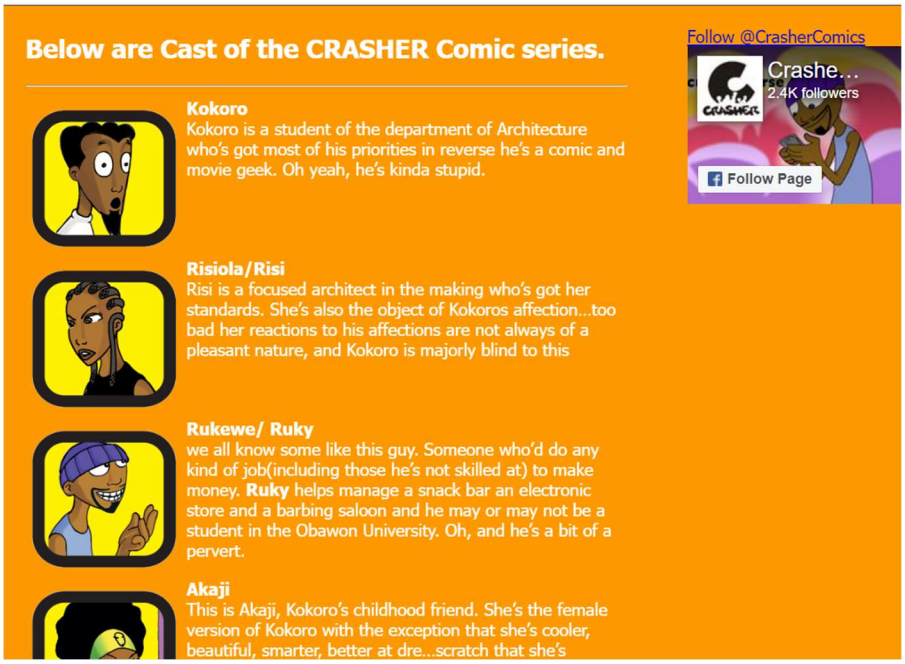

5/5/24, 2:49 PM
Alternative Shared Kitchen
What can you change in the kitchen and make in new ways?
10 responses
The kitchen cabinets
Creation of waste chutes, easier waste separation mechnisms and more effective smoke and odour extractors.
Adding more automated machines
Automate cooking methodology to save time and become more ease
Providing exit door
The food cooking itself
Baking
Nothing
I've never stop to think it about it
The compartments for the rubbish.


5/5/24, 2:49 PM
Alternative Shared Kitchen
Where do you get your recipes?
11 responses
Usually online, I search for a specific recipes and I follow related cooking pages and channels on Instagram and YouTube.
Online
From malls and open markets
Online and from my mother
Grocery store like Willys
Willys
From the local markets
IG , YouTube
Shops
From my family
From my family


5/5/24, 2:49 PM
Alternative Shared Kitchen
How can you enhance the **functionality** of the kitchen?
10 responses
By having multi-functional utensils or kitchen stuffs to save some spaces.
By designing a functional one that meets the intended purpose while exemplifying aesthetics at its peak
By integrating social features at the design stage.
By adding more equipment which would make it easier to work in parallel rather than waiting for others to finish there meals
Having all the modern equipment and household cooking gadgets
By having an exit door
Have it more equipped
A space where people can have enough space to have meetings
Optimitzating the process
Removing unnecessary staff or keeping it to a minimum.

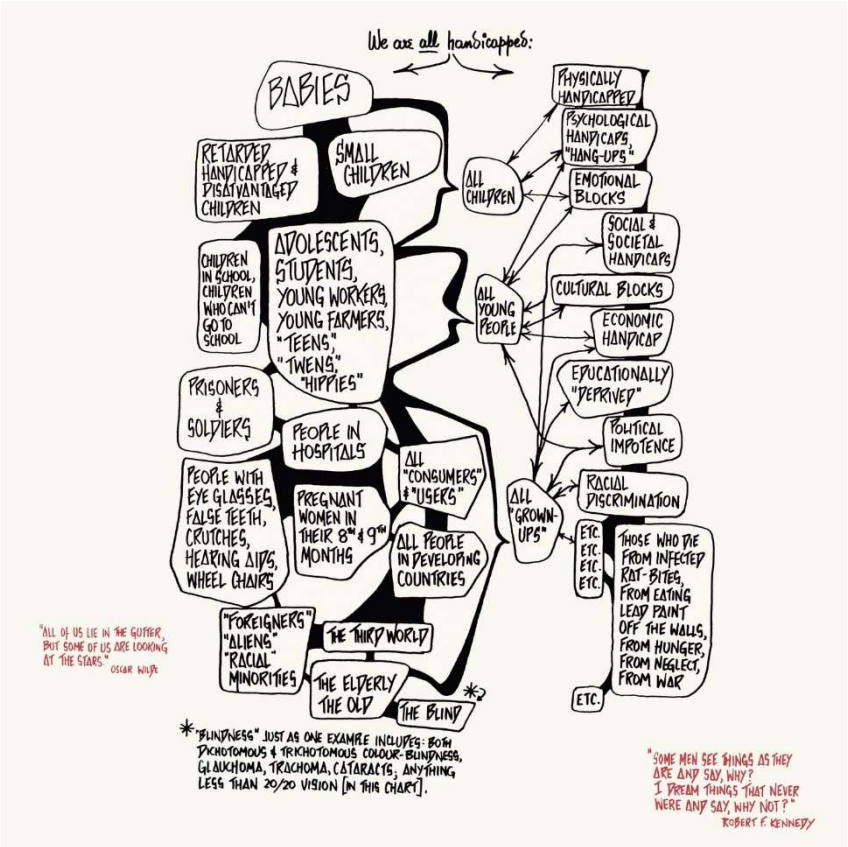

5/5/24, 2:49 PM
Alternative Shared Kitchen
What don't you like about the kitchen?
11 responses
Lack of drawers or cabinets. Also top shelves would be harder to access. Sometimes things get behind a buch of stuff and may be forgotten. My current kitchen has no top board cabinet counter and it is challenging to manage around when I need to prepare the food.
Poor Ventilation
Loneliness
I don't like sharing kitchen with strangers that i do not know very well
Being close to the living room. A traditional new design in the Europe I say.
Not adequately ventilated
I dont like cooking
The fridge is small and table top too
The space
The fact that I've to cook everytime
The extractor hood

Appendix 2



Risi's character in Crasher comic

Appendix 3



Victor Papanek's Design for the Real World (1968) similar to my thesis persona – "Different", (Vitra Design Museum, 2019)